

ELASTICITIES, MARGINS AND PROFITABILITY

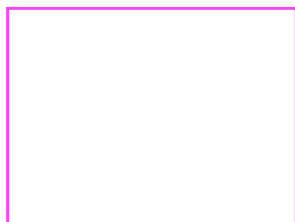
Overview

2

Last lecture:

- ➔ *Optimal pricing* requires knowledge of **demand**
- ➔ Shape of demand determines pricing
 - **Trade-off**: sales volume vs margin
- ➔ We used **EVC** as coarse approximation of demand

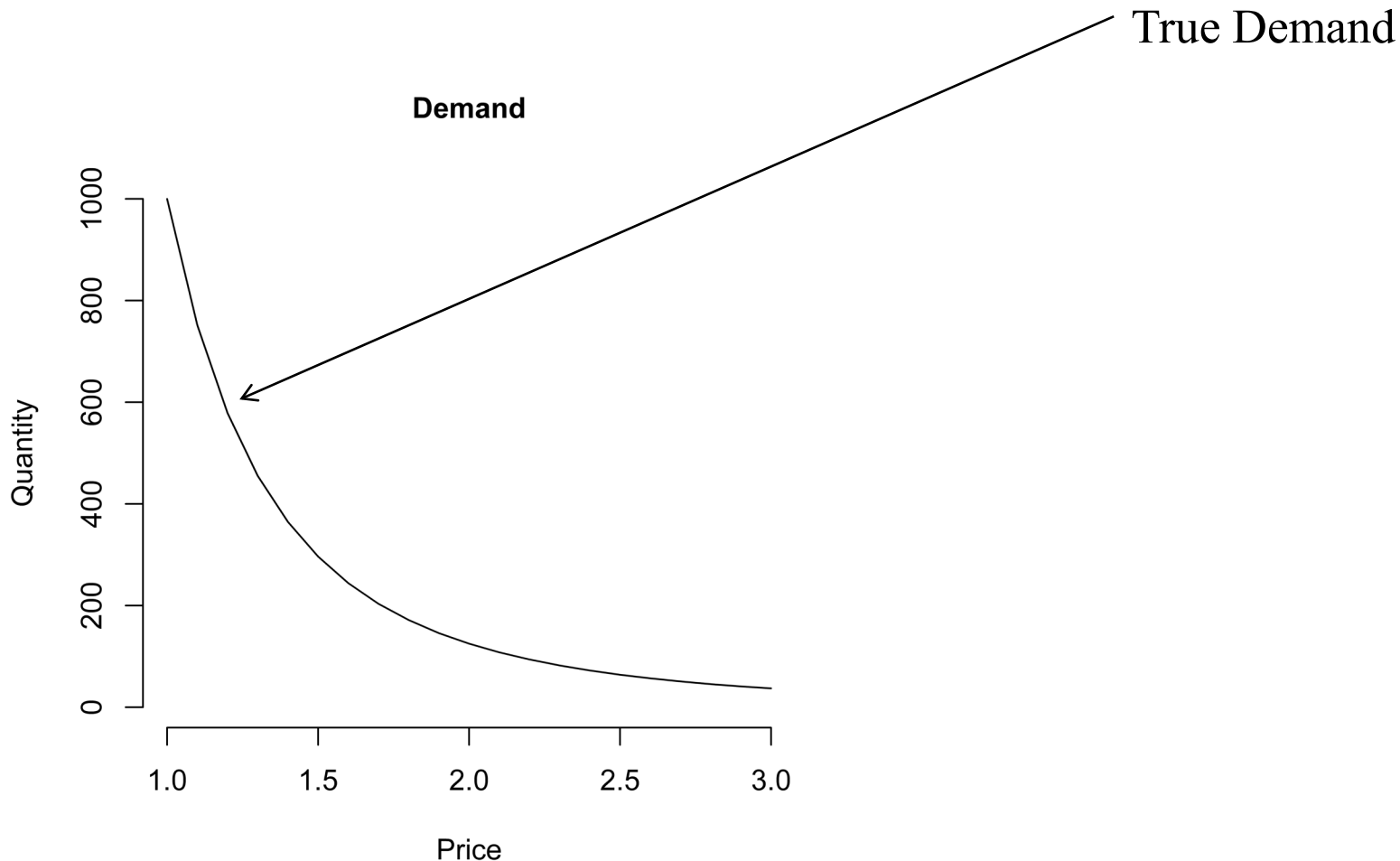
This lecture:



- ➔ With better data, can use a *smooth* model of demand
- ➔ Measure the **price elasticity** of demand
- ➔ Use price elasticity to set **optimal uniform price**
- ➔ optimal **uniform price** as **GROWTH!!!**

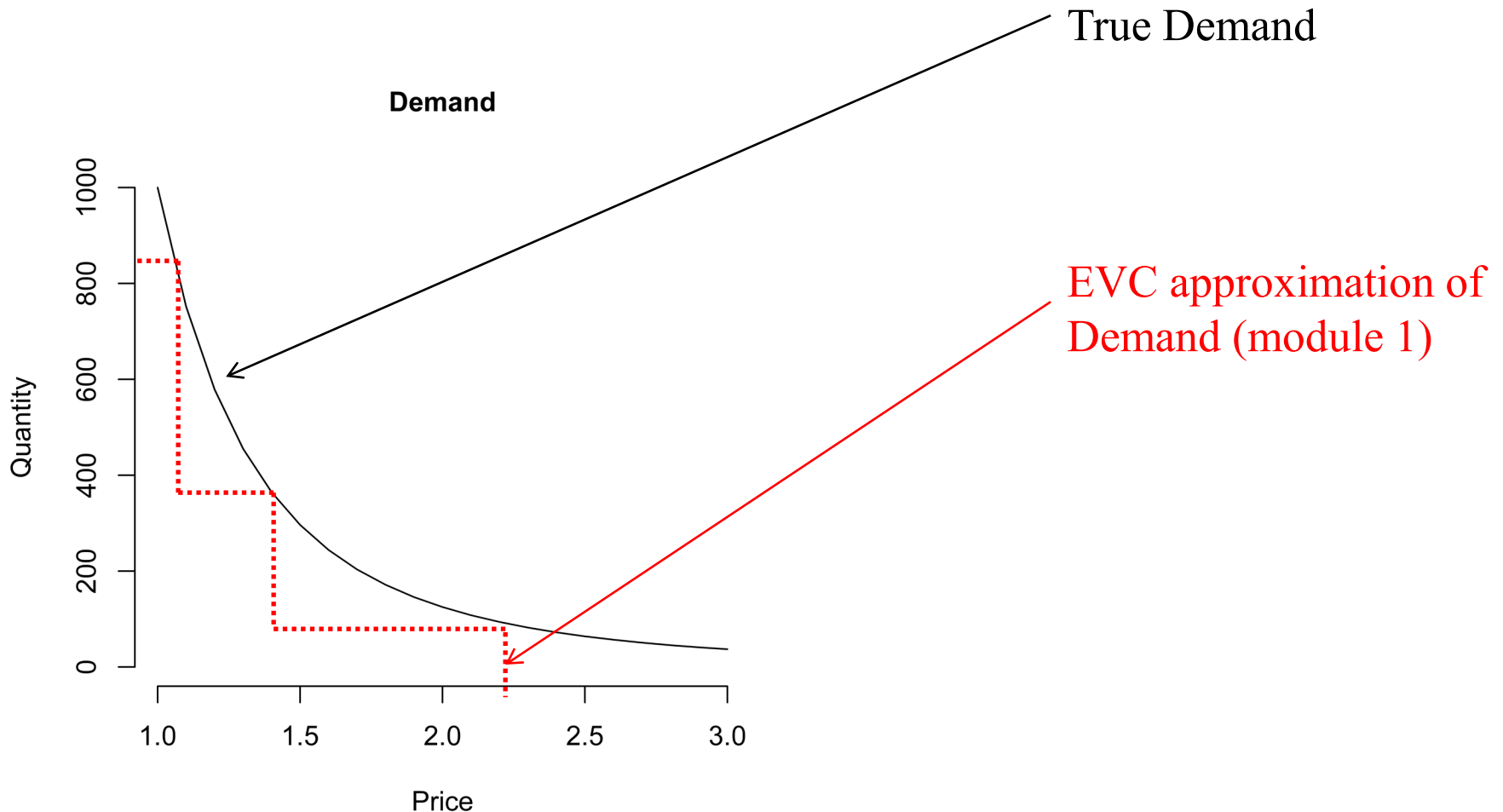
Recall: EVC was used to approximate the Demand Curve

3



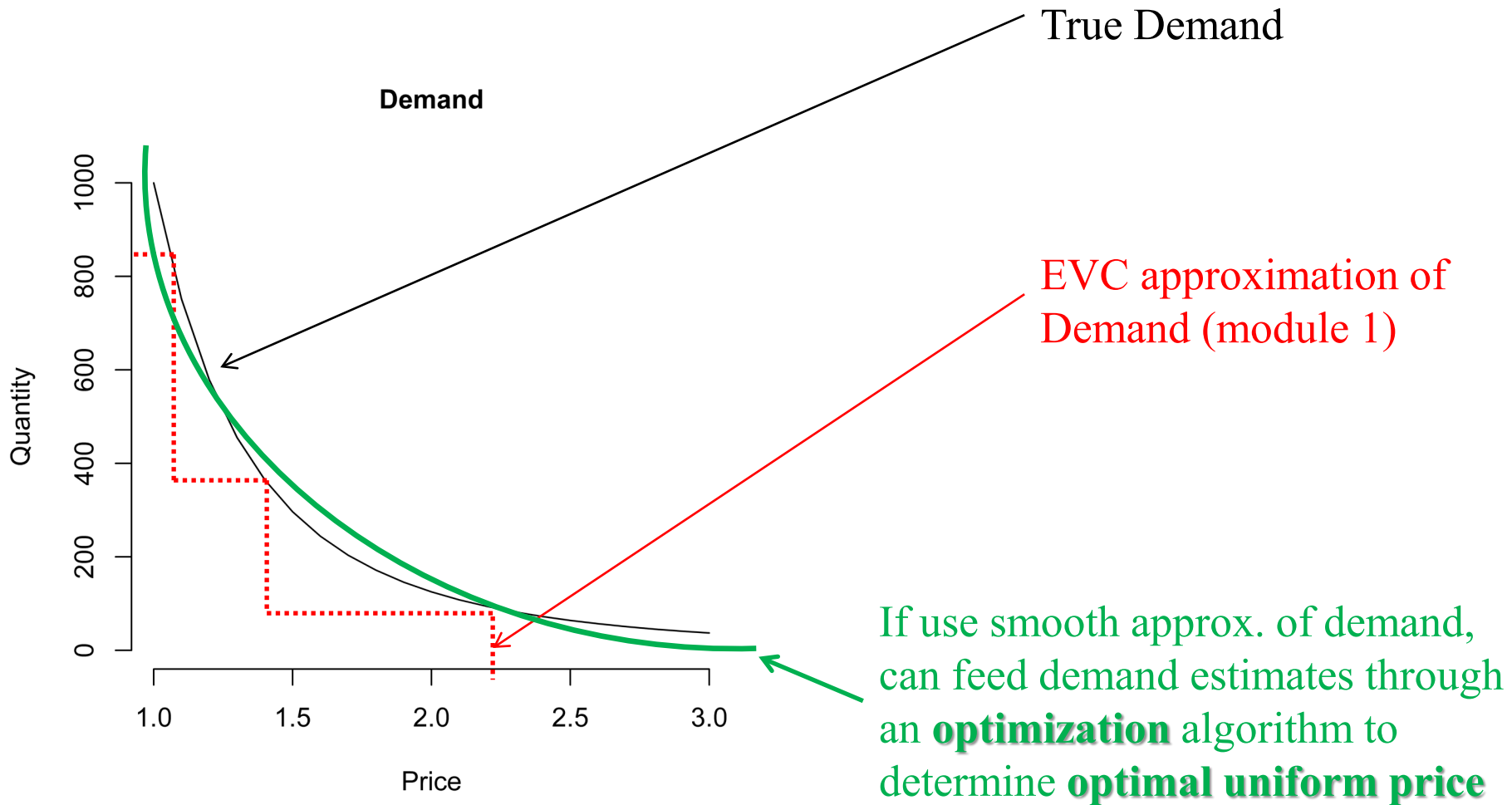
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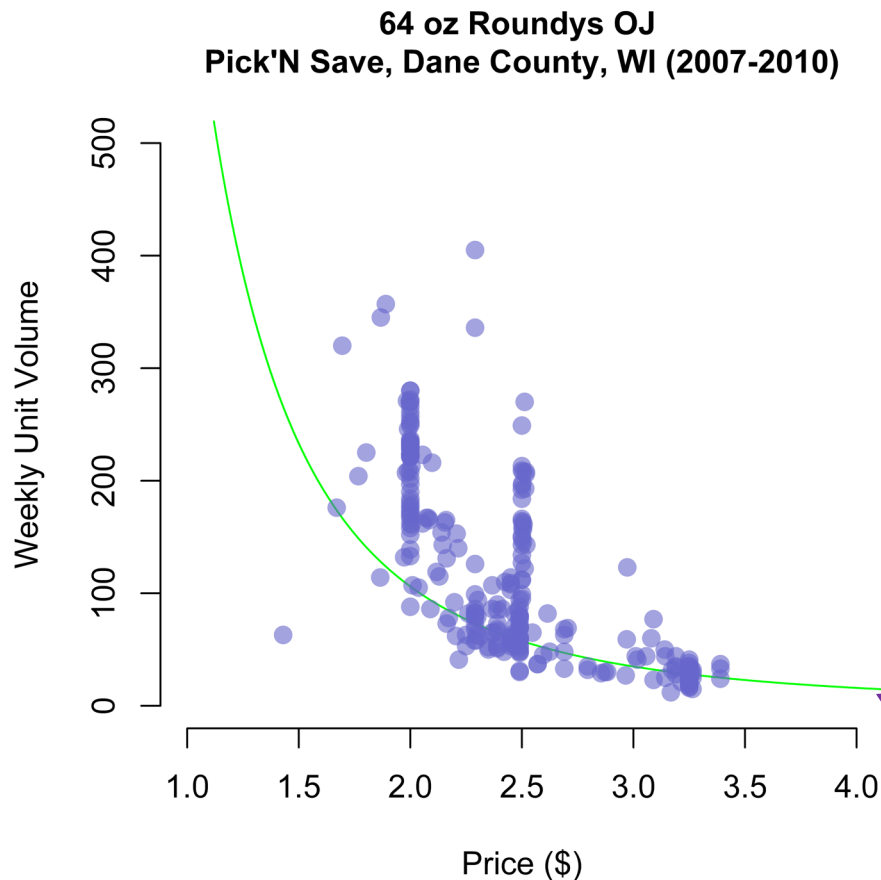


Field Data and Regression

Example: Demand for 64 oz Roundy's OJ



4



Regression estimate of demand
(see *Data-Driven Marketing*
course & extra slides)

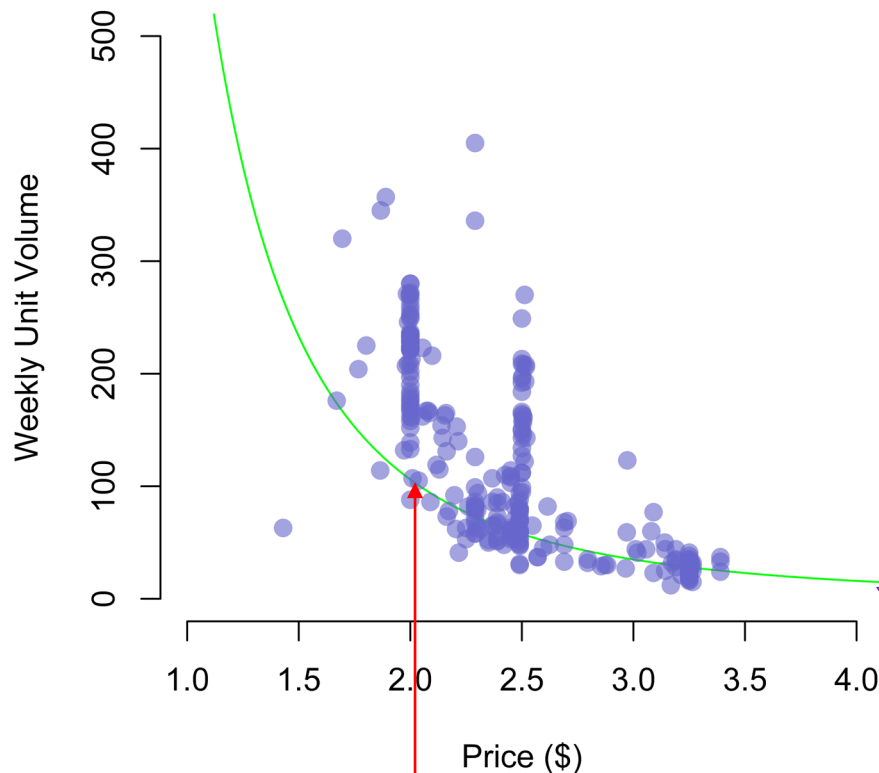
Field Data and Regression

Example: Demand for 64 oz Roundy's OJ



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64 oz Roundys OJ
Pick'N Save, Dane County, WI (2007-2010)



Shelf Price of \$2

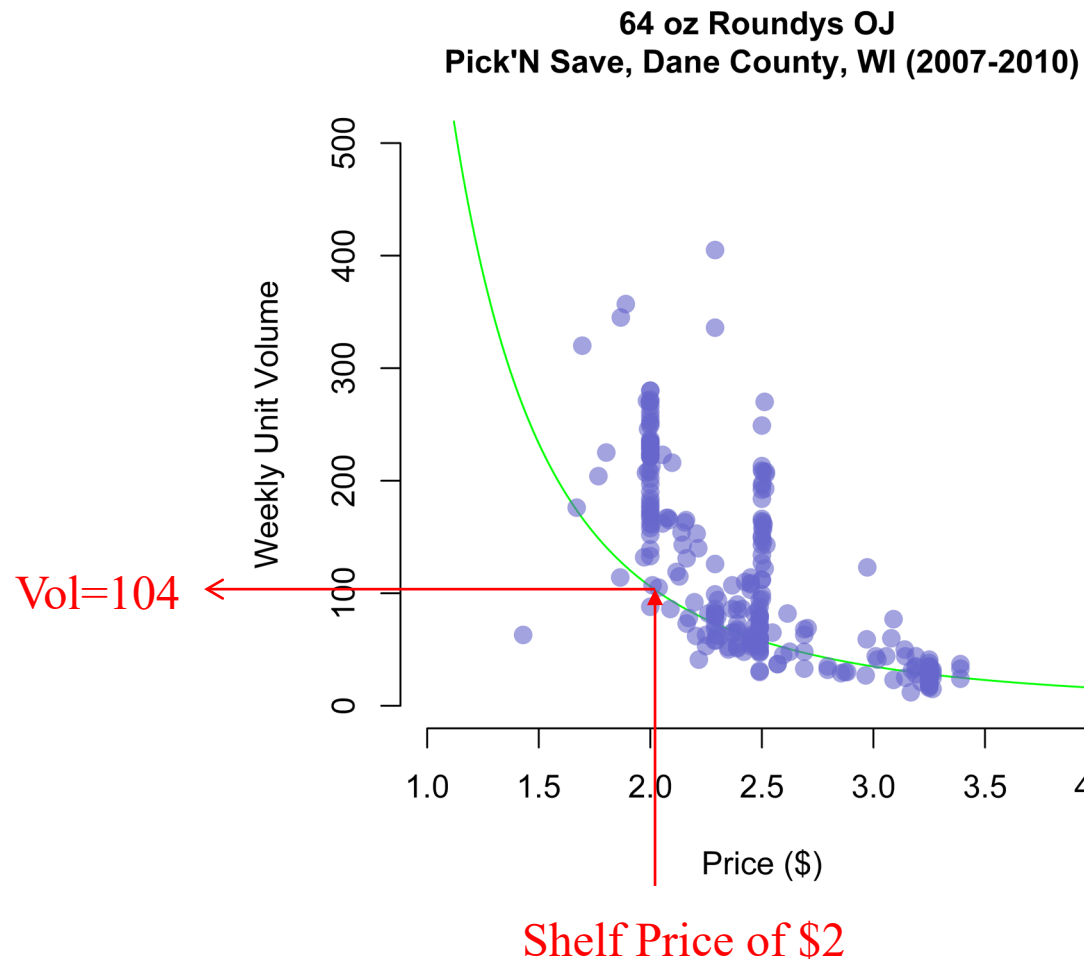
Regression estimate of demand
(see *Data-Driven Marketing*
course & extra slides)

Field Data and Regression

Example: Demand for 64 oz Roundy's OJ



4



Weekly data plots of sales vs price.

Regression estimate of demand
(see *Data-Driven Marketing*
course & extra slides)

Review of price elasticity of demand

5

- ▣ Measures price sensitivity of the demand curve *at a specific point*

$$\beta = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} \times \frac{100\%}{100\%}$$

See Appendix for discussion of typical magnitudes of estimated elasticities in practice

Review of price elasticity of demand

5

- ▣ Measures price sensitivity of the demand curve *at a specific point*
- ▣ Unit-less Measure
- ▣ For more details, see handout on demand.

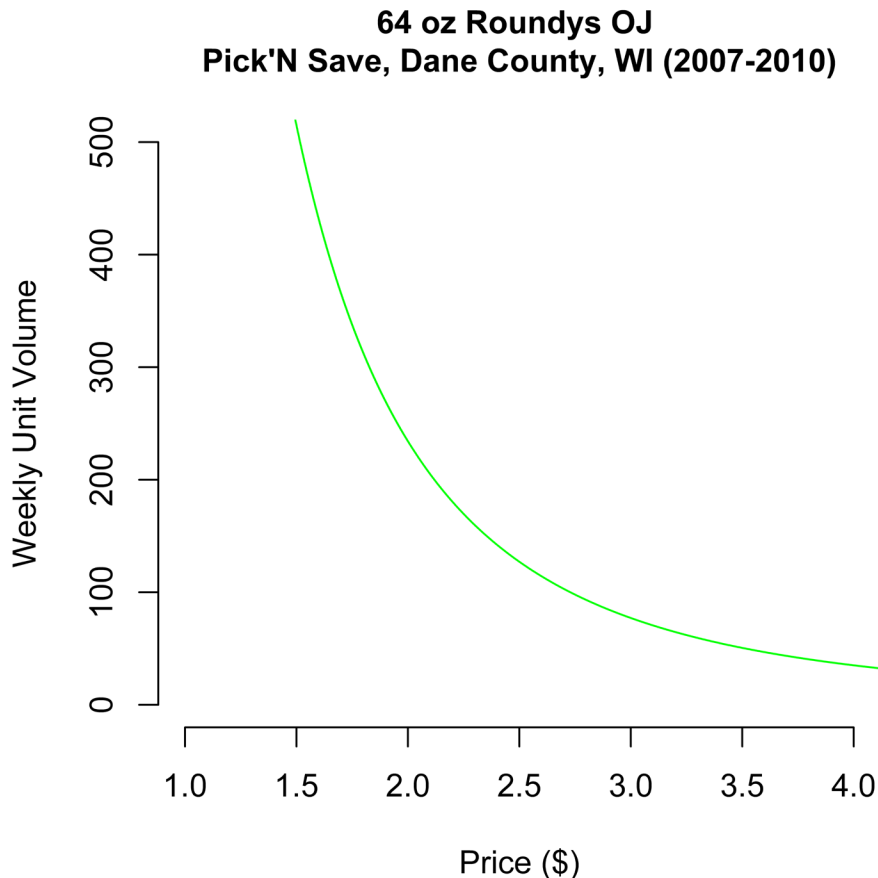
$$\begin{aligned}\beta &= \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} \times \frac{100\%}{100\%} \\ &= \frac{\% \Delta Q}{\% \Delta P}\end{aligned}$$

Demand Elasticity at Pick N' Save

Example: 64 Oz Roundy's Orange Juice



6



Demand Elasticity at Pick N' Save

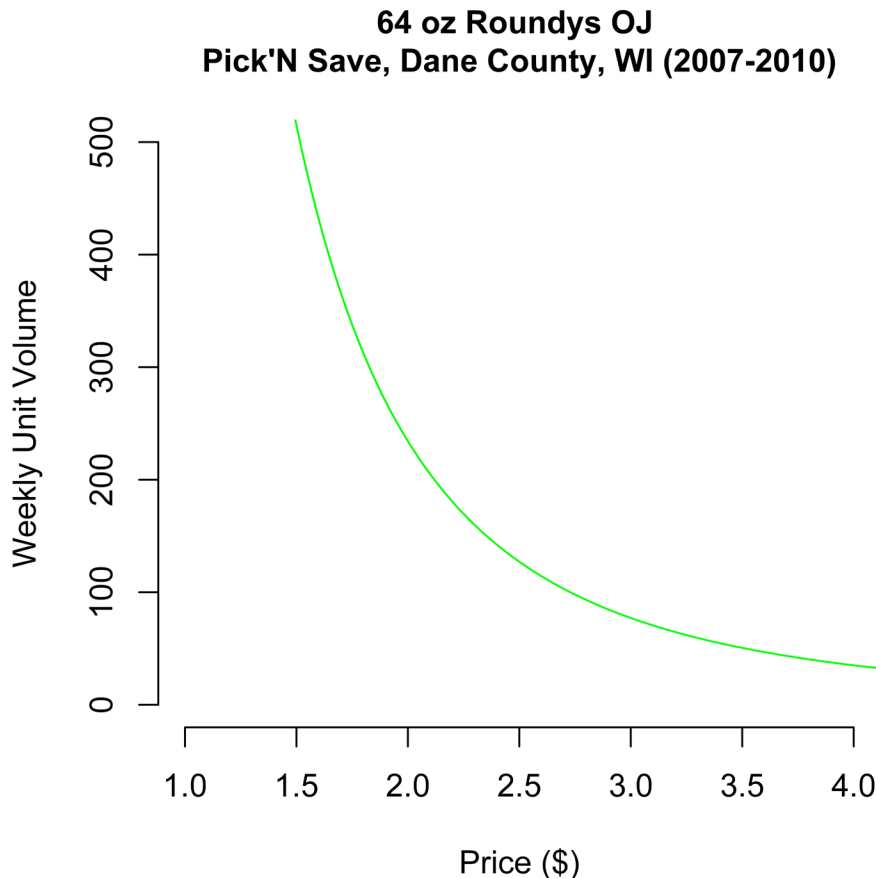
Example: 64 Oz Roundy's Orange Juice



6

- Regression analysis of demand data gives us:

$$\beta = -3.11$$



Demand Elasticity at Pick N' Save

Example: 64 Oz Roundy's Orange Juice



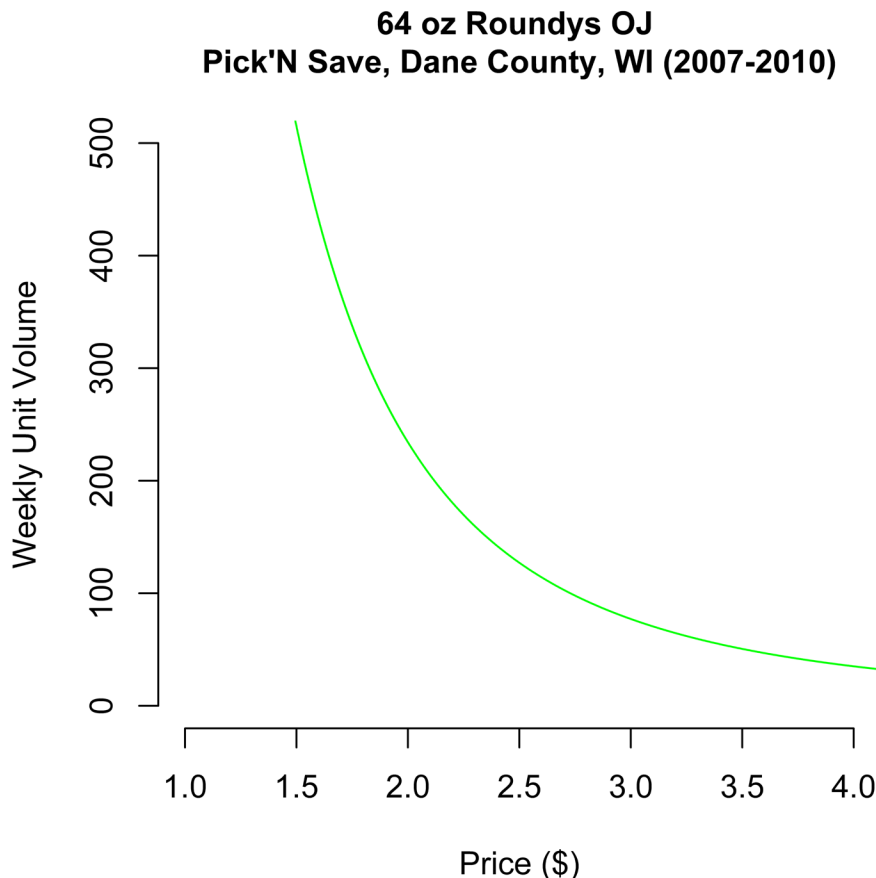
6

- Regression analysis of demand data gives us:

$$\beta = -3.11$$

➔ *a 1% price cut generates a 3.1% volume increase*

(Loosely, average elasticity in range of data)



Monopoly price and elasticity

The Profit-Maximizing **Uniform Price**

7

- Assume
 - ▣ Single product firm
 - ▣ No competitive response
 - ▣ **Business Objective: maximize profits**

Set price (p) to maximize variable profit:

$$(p - c) \times Q(p)$$

Monopoly price and elasticity

The Profit-Maximizing **Uniform Price**

7

- Assume
 - ▣ Single product firm
 - ▣ No competitive response
 - ▣ **Business Objective:** maximize profits

Set price (p) to maximize variable profit:

$$(p - c) \times Q(p)$$

Method.
Recall; Take derivative of
this equation with respect
to P , and then set to zero.

Unlike last lecture, now assume $Q(P)$ {i.e., demand} is smooth and differentiable so that we can compute elasticity

Monopoly price and elasticity

The inverse-elasticity rule

8

- **Optimal Price** is set such that:

$$p^* = \left(\frac{\beta}{1 + \beta} \right) \times c$$

OR

inverse-elasticity rule

$$\frac{p^* - c}{p^*} = -\frac{1}{\beta}$$

When i maximise profit, the contribution to my profit is 1 over elasticity.

The higher beta (more sensitive to price, meaning i have low EVC), the less profits i will make.

With low beta below 1 (low price sensitivity), right side is large, meaning (p-c), aka profit is high as a proportion of price.

Monopoly price and elasticity

The inverse-elasticity rule

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(see math Appendix for derivation)

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- But this is a useful starting point

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inverse-elasticity rule

(see math Appendix for derivation)

- In practice, most firms not monopolists & competitors not passive
- But this is a useful starting point
- Note: elasticity usually a function, i.e., $\beta(p)$ not a number
(see Ziprecruiter appendix)

Pricing on the elastic region of demand

9

- **Claim: if a firm maximizes its (current) profits, then**

$$|\beta| \geq 1$$

Pricing on the elastic region of demand

9

- **Claim: if a firm maximizes its (current) profits, then**

$$|\beta| \geq 1$$

If our beta is less than 1, it means that we have not exercised our pricing power well enough.

If we know our beta is accurate, we use the pricing power formula to set the prices (page 20).

Therefore, you should raise your price if $|\beta| < 1$

Can you see why?

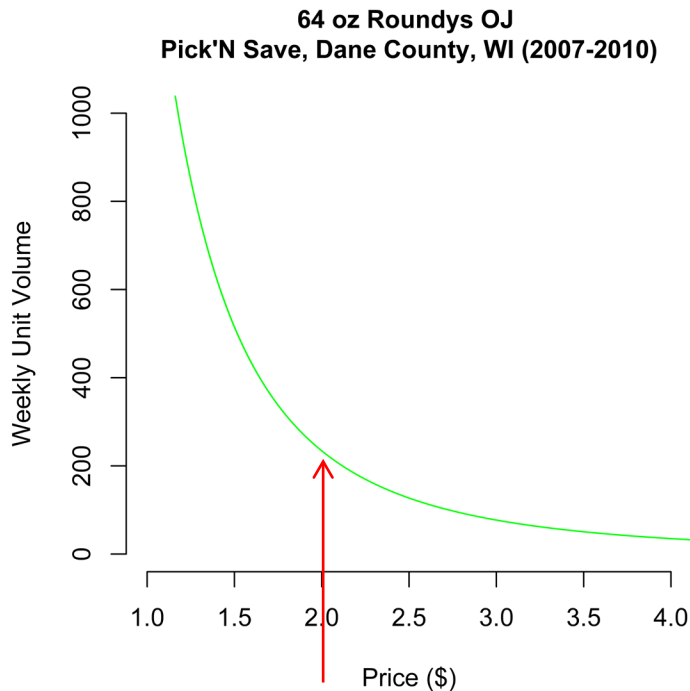
(hint: play with inverse-elasticity rule)

- Be careful. Very inelastic demand estimates might also reflect a problem with your data or your method used to estimate β .

Demand regressions at Pick N' Save

Example: Refrigerated Orange Juice

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Current price \$2.00



Current price \$2.99

Optimal Prices at Pick N' Save

Example: Refrigerated Orange Juice

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- Let's design a ***pricing dashboard*** for refrigerated orange juice
- The dashboard is designed to help with decision-making
- The dashboard will consist of:
 - ▣ metrics
 - ▣ visualizations

Teach the team how to build a pricing dashboard

A Pricing Dashboard for Pick N' Save

Example: Refrigerated Orange Juice

12

	Own-Elasticity
Roundy's 64 Oz	-3.11
Minute Maid 64 Oz	-2.40



Wholesale prices

Minute Maid 64 Oz = \$2.06

Roundy's 64 Oz = \$1.67

Note: we are ignoring cannibalization! We will consider product line pricing issues later in course.

A Pricing Dashboard for Pick N' Save

Example: Refrigerated Orange Juice

12

	Own-Elasticity	Optimal Margin
Roundy's 64 Oz	-3.11	32%
Minute Maid 64 Oz	-2.40	42%

β

$-\frac{1}{\beta}$

Wholesale prices

Minute Maid 64 Oz = \$2.06

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A Pricing Dashboard for Pick N' Save

Example: Refrigerated Orange Juice

12

	Own-Elasticity	Optimal Margin	Optimal Mark-up
Roundy's 64 Oz	-3.11	32%	47.39%
Minute Maid 64 Oz	-2.40	42%	71.6%

$$\beta$$


$$-\frac{1}{\beta}$$


$$\left(\frac{\beta}{1+\beta} - 1\right) \times 100$$


Wholesale prices

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A Pricing Dashboard for Pick N' Save

Example: Refrigerated Orange Juice

12

	Own-Elasticity	Optimal Margin	Optimal Mark-up	Optimal Price
Roundy's 64 Oz	-3.11	32%	47.39%	\$2.46
Minute Maid 64 Oz	-2.40	42%	71.6%	\$3.53

β

$-\frac{1}{\beta}$

$\left(\frac{\beta}{1+\beta} - 1\right) \times 100$

Wholesale prices

Minute Maid 64 Oz = \$2.06

Roundy's 64 Oz = \$1.67

Optimal price = wholesale price x optimal markup.

$\frac{\beta}{1+\beta} \times (\text{unit cost})$

Note: we are ignoring cannibalization! We will consider product line pricing issues later in course.

A Pricing Dashboard for Pick N' Save

Example: Refrigerated Orange Juice

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$$\beta$$

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Wholesale prices

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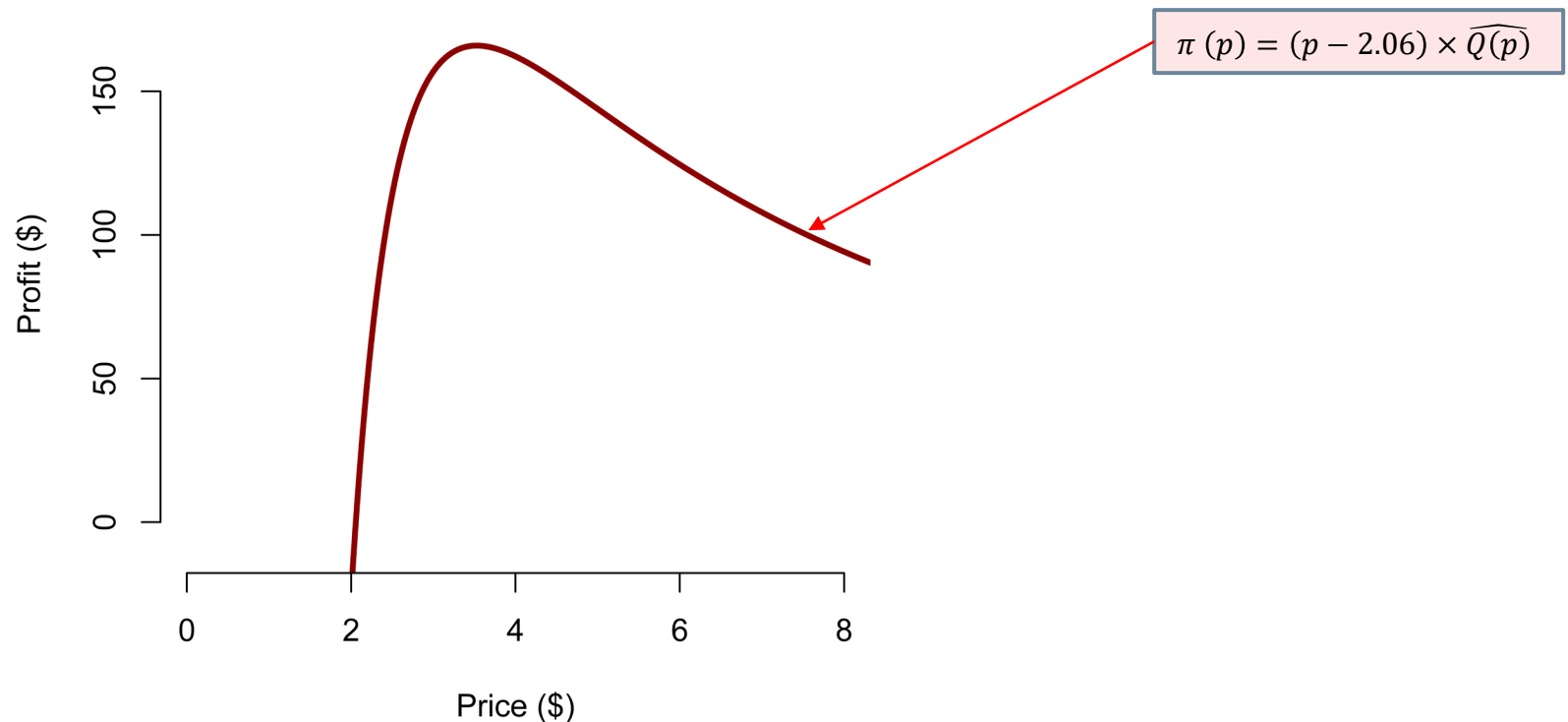
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Profitability Visualization

Example: Refrigerated Orange Juice

13

64 oz Minute Maid OJ
Pick'N Save, Dane County, WI (2007-2010)

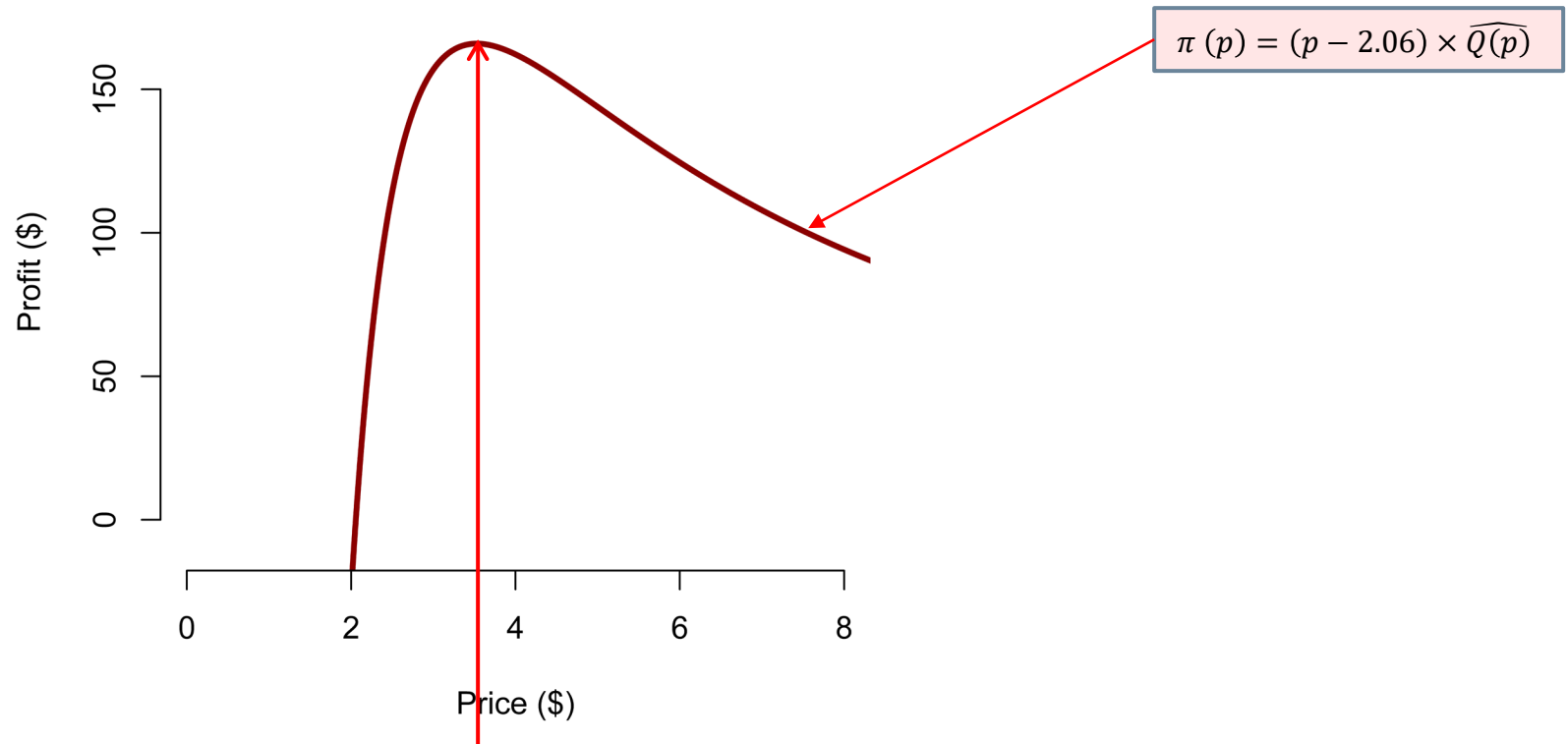


Profitability Visualization

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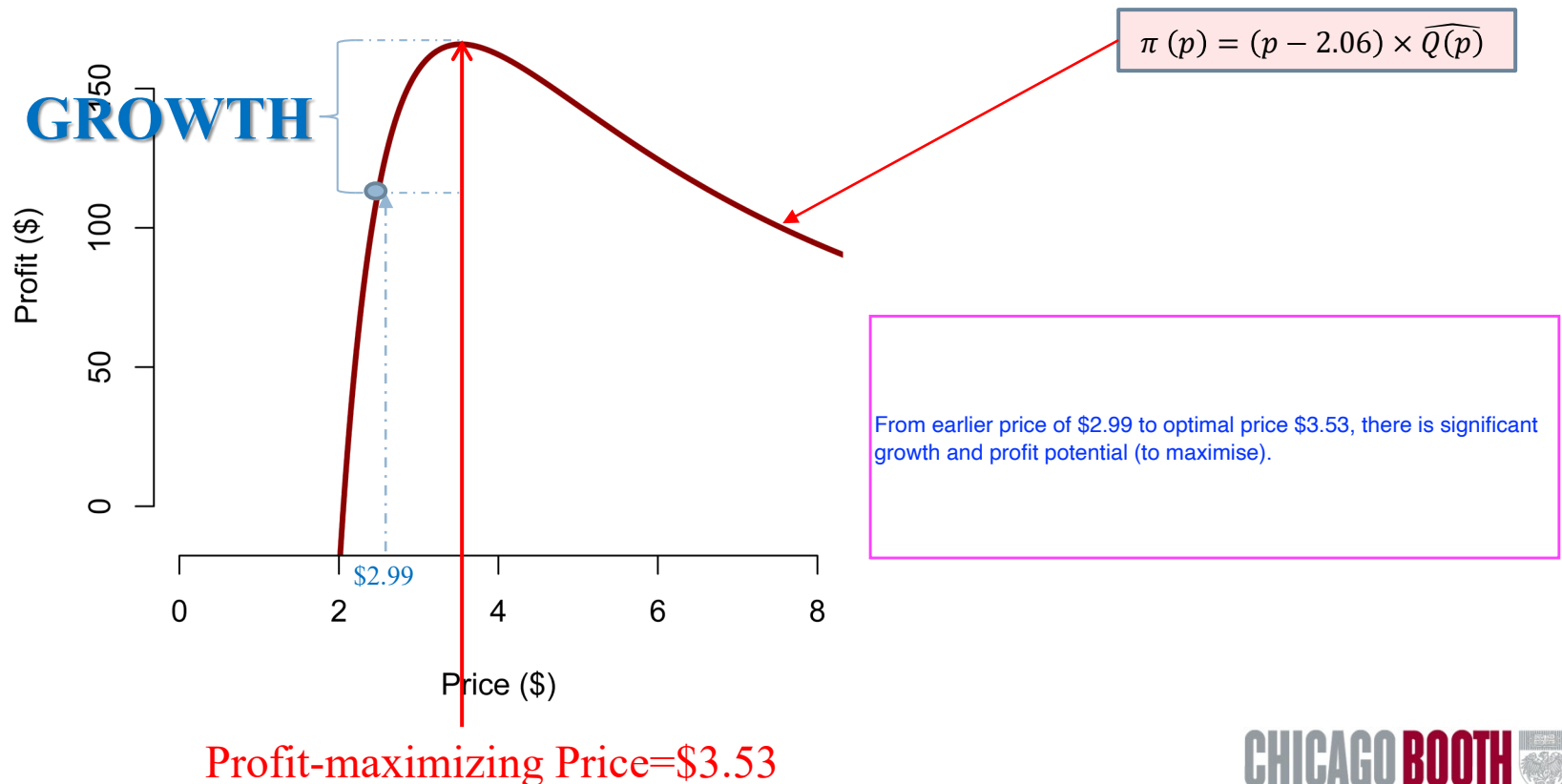
Profit-maximizing Price=\$3.53

Profitability Visualization

Example: Refrigerated Orange Juice

13

64 oz Minute Maid OJ
Pick'N Save, Dane County, WI (2007-2010)



Measuring the Price Elasticity

Field Experiments

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- A simple field experiment can be used to measure the own-price elasticity (e.g., *A-B testing*)
- Randomly assign subjects into treatment and control conditions
e.g., subjects could be customers, stores, cities etc.
- Assign different prices to treatment and control conditions
- Apply elasticity formula to manipulated “% changes in prices” and observed “% changes in sales”

$$\beta = \frac{\% \Delta Q}{\% \Delta P}$$

Measuring the Price Elasticity

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$$\beta = \frac{\% \Delta Q}{\% \Delta P} \leftarrow \text{Experimental design}$$

Measuring the Price Elasticity

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← Experimental outcome

← Experimental design

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Pricing at Ziprecruiter



ZipRecruiter[®]

Pricing at Ziprecruiter.com

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- Academic collaboration with Ziprecruiter.com
- Fastest growing HR company: 75,000 employers per month
- Online matching platform for recruiters and applicants
- High conversion rates and high quality matches
- August 2015: charging \$99 per month to “starter” businesses

Pricing at Ziprecruiter.com

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... but why \$99?

A Price Experiment at Ziprecruiter.com

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- How do you evaluate the \$99 price?
- How do you measure **demand** for a B2B digital service?
- **method:**

We ran a **B2B price experiment** to measure **demand** and determine the optimal price at Ziprecruiter...

The Ziprecruiter Price Experiment

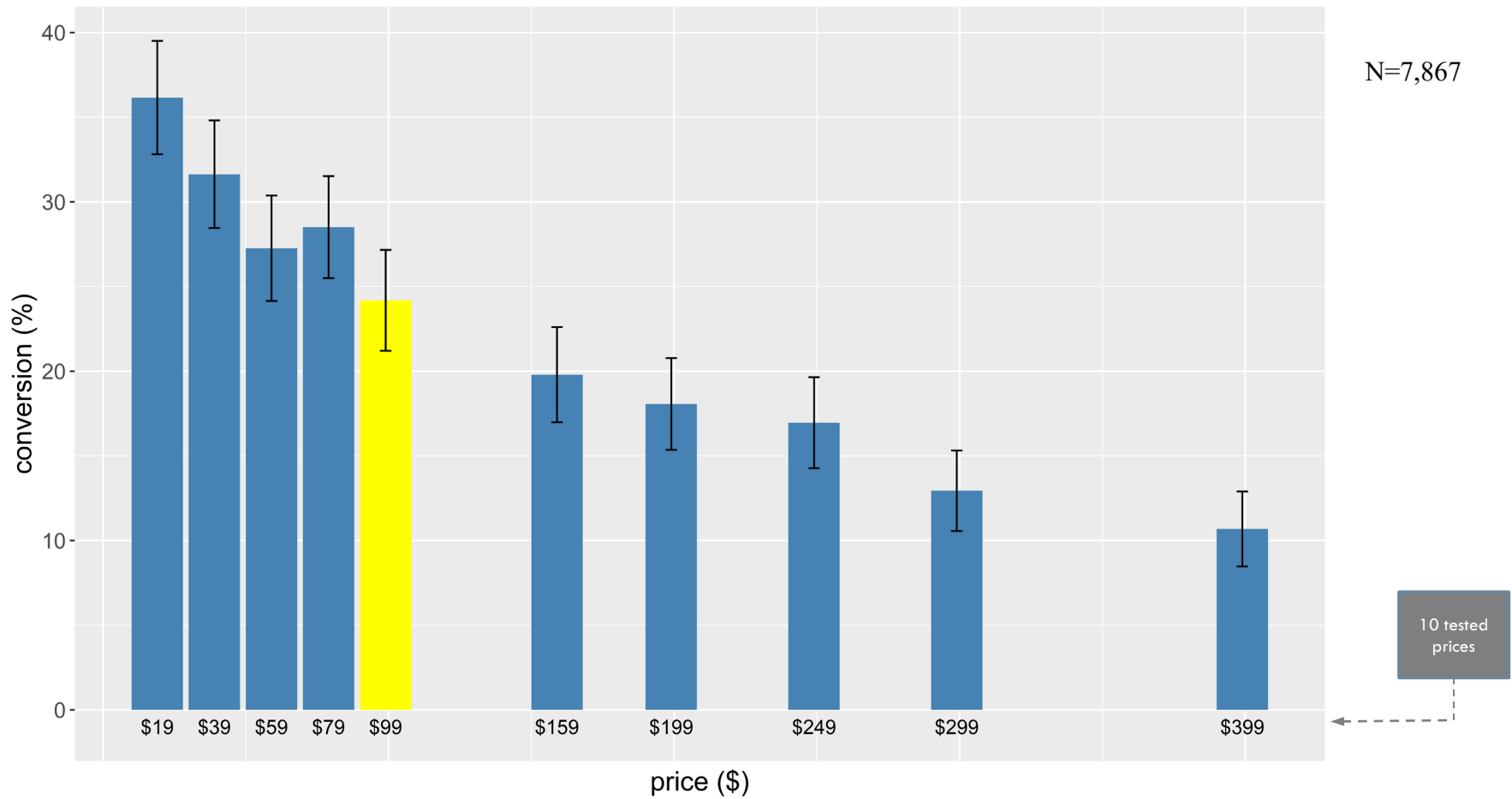
Method

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- 10 prices tested:
 $\{\$19, \$39, \$59, \$79, \$99, \$159, \$199, \$249, \$299, \$399\}$
- All new customers in September 2015
 - ▣ Customer had to register and advance to paywall
 - ▣ Each customer assigned randomly to a price
 - ▣ Track whether customer submitted credit card information and was billed for a month of service
- Some concern about a “*price cliff*” above \$99 (see Appendix)
 - ▣ This is testable too...

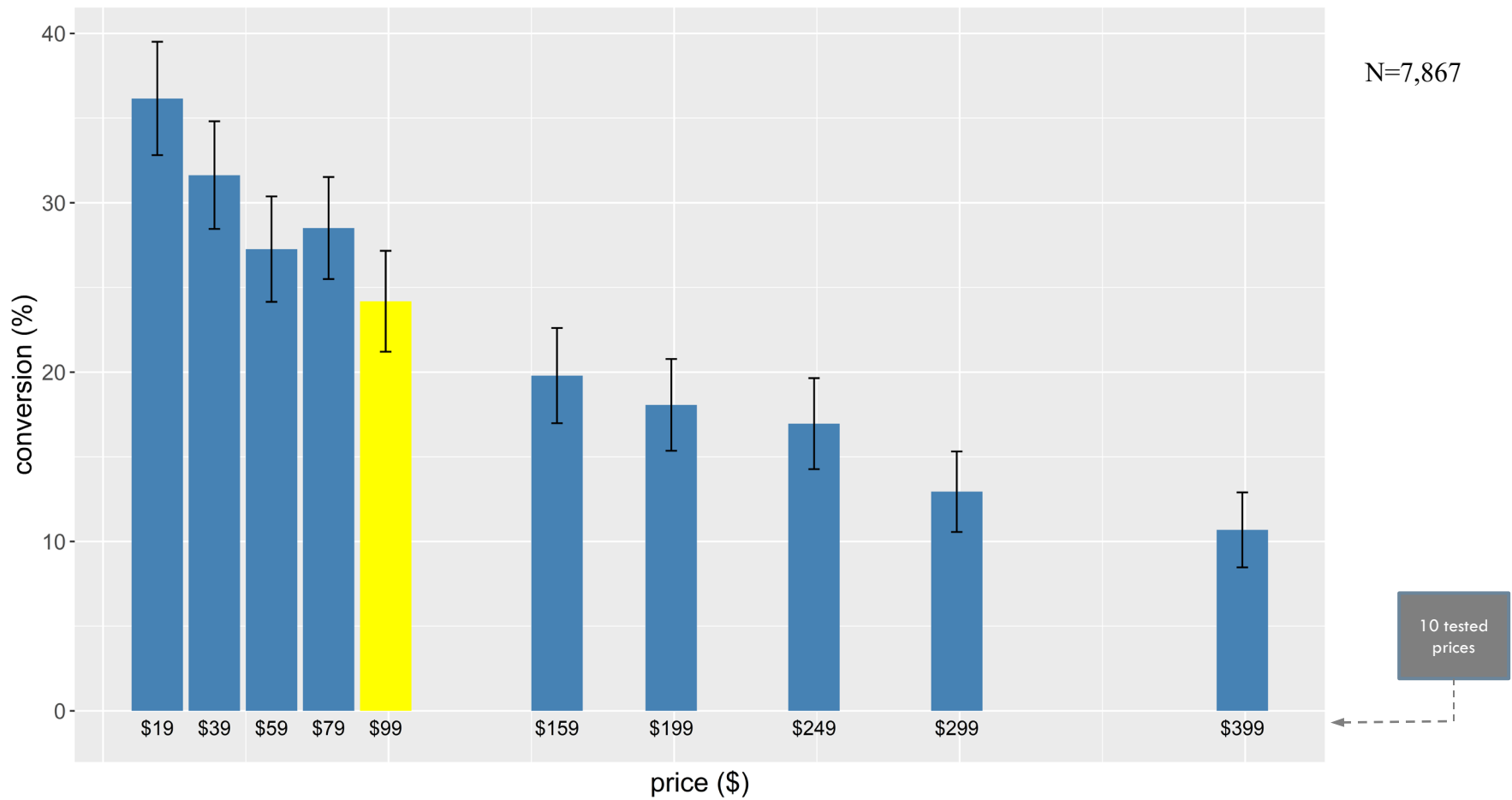
The Ziprecruiter Price Experiment

Price & Conversion



The Ziprecruiter Price Experiment

Price & Conversion

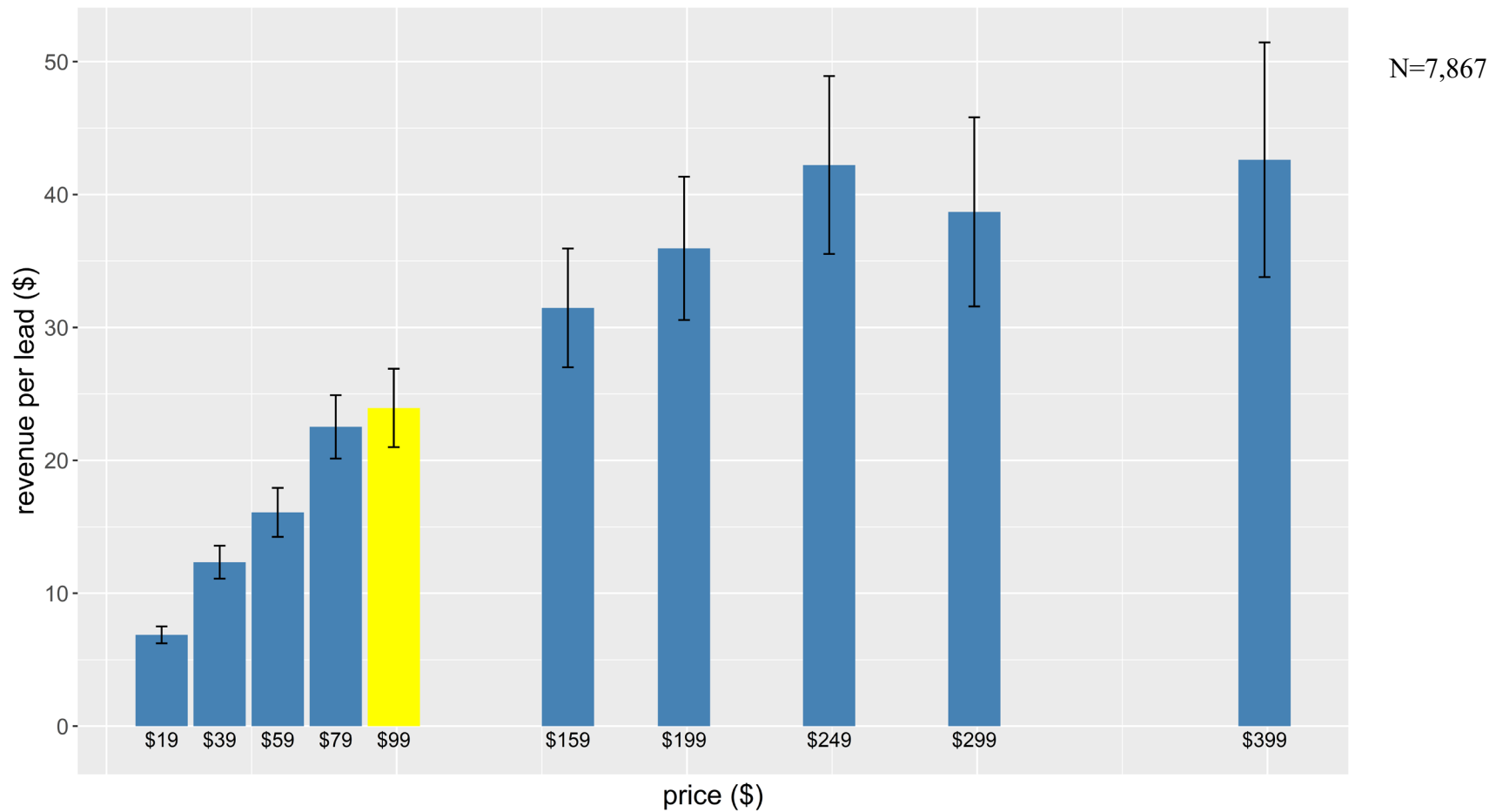


□ Do not lose much business even with 4-fold price increase!

The Ziprecruiter Price Experiment

Implementation: Price & revenue per lead

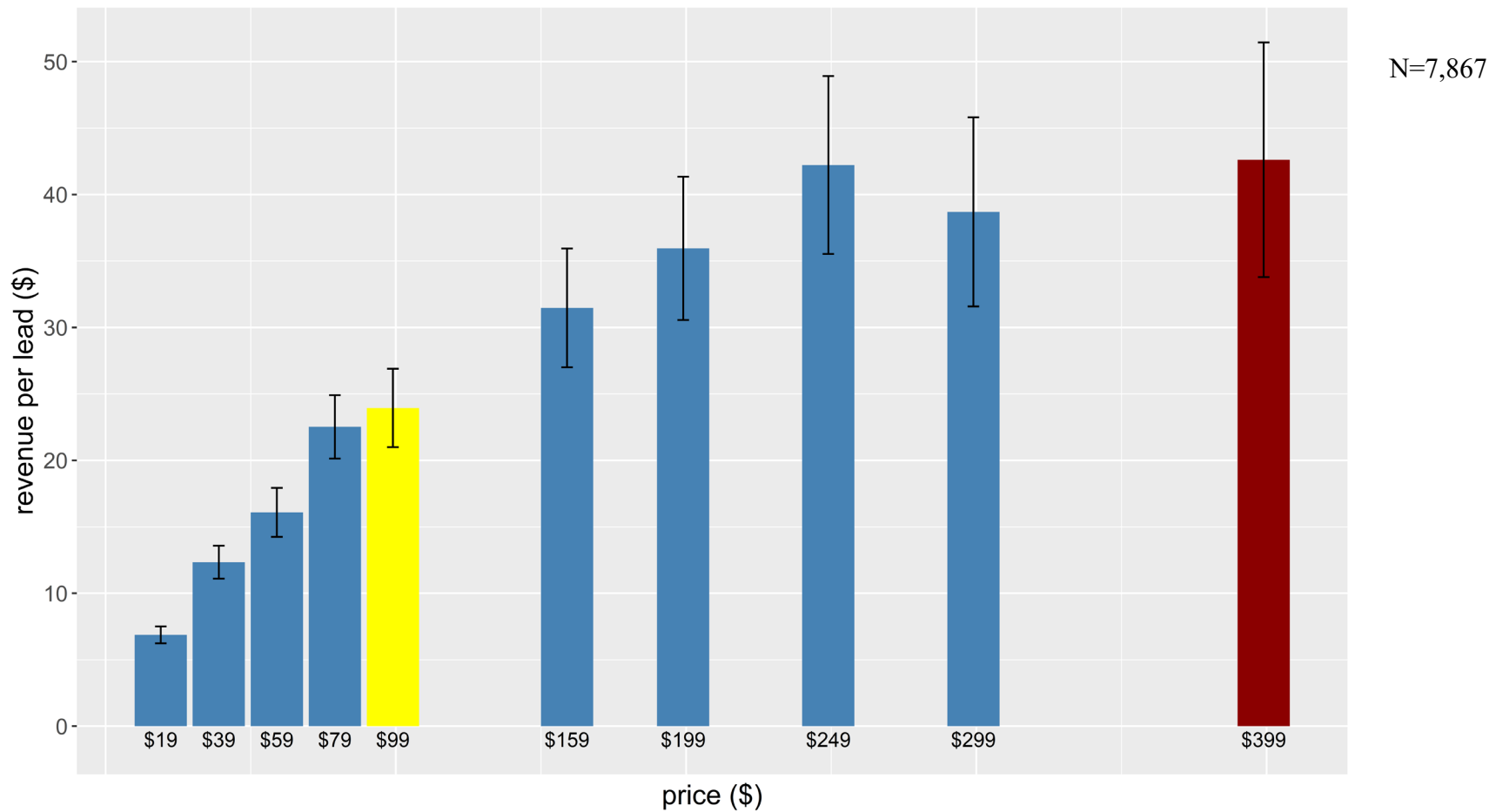
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The Ziprecruiter Price Experiment

Implementation: Price & revenue per lead

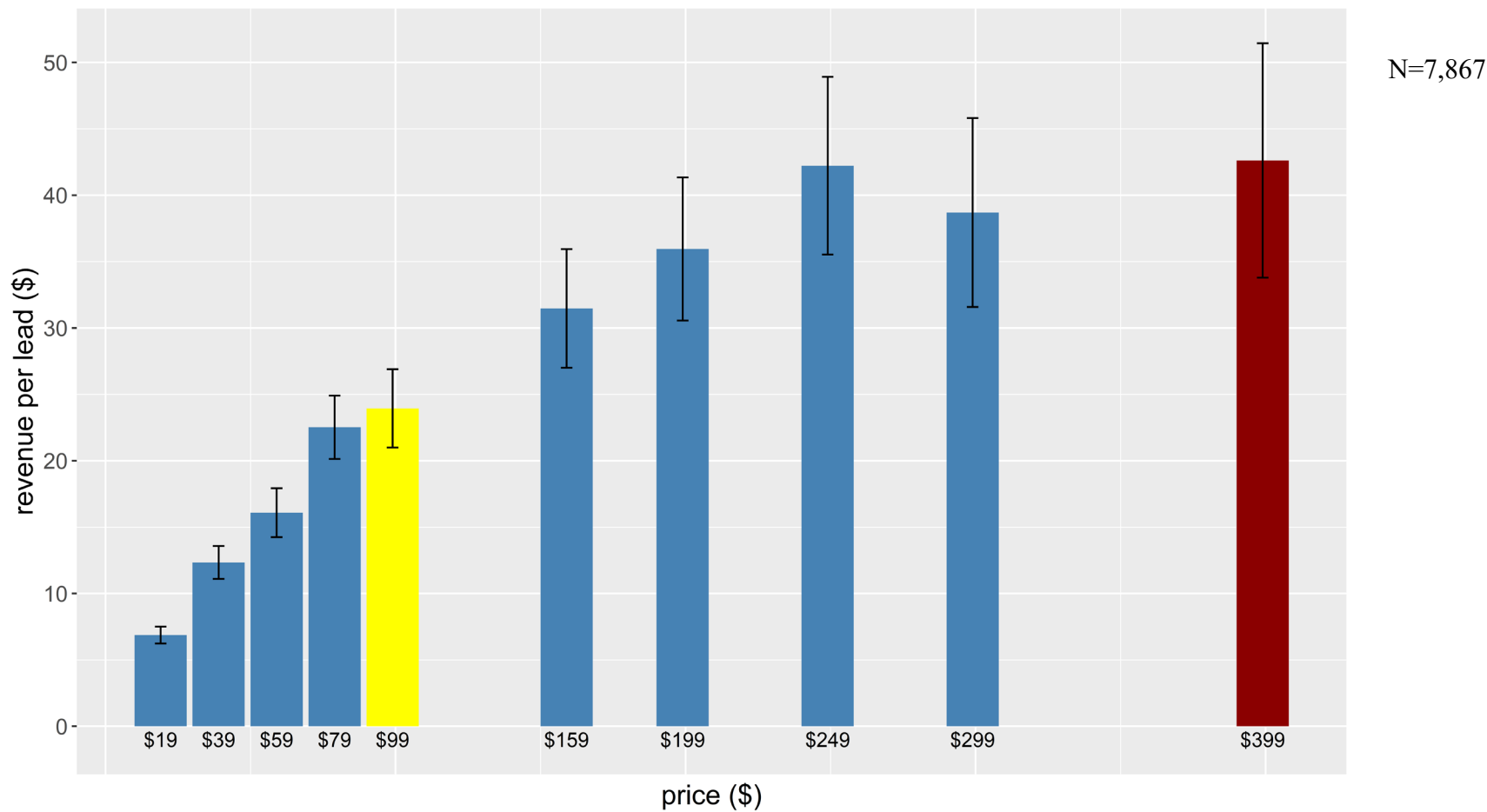
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The Ziprecruiter Price Experiment

Implementation: Price & revenue per lead

21

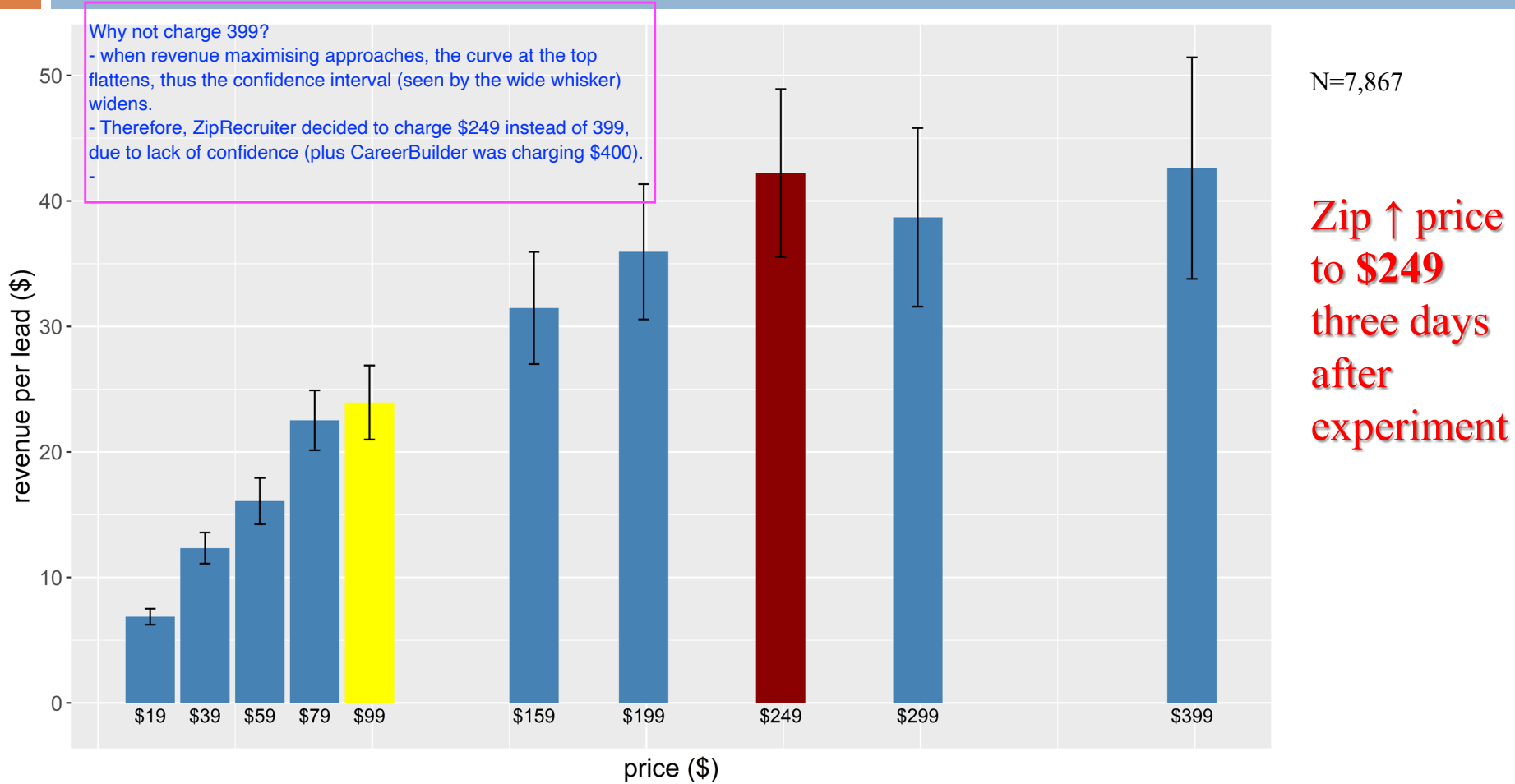


□ Optimal price on grid increases profits by over 60%!!!

The Ziprecruiter Price Experiment

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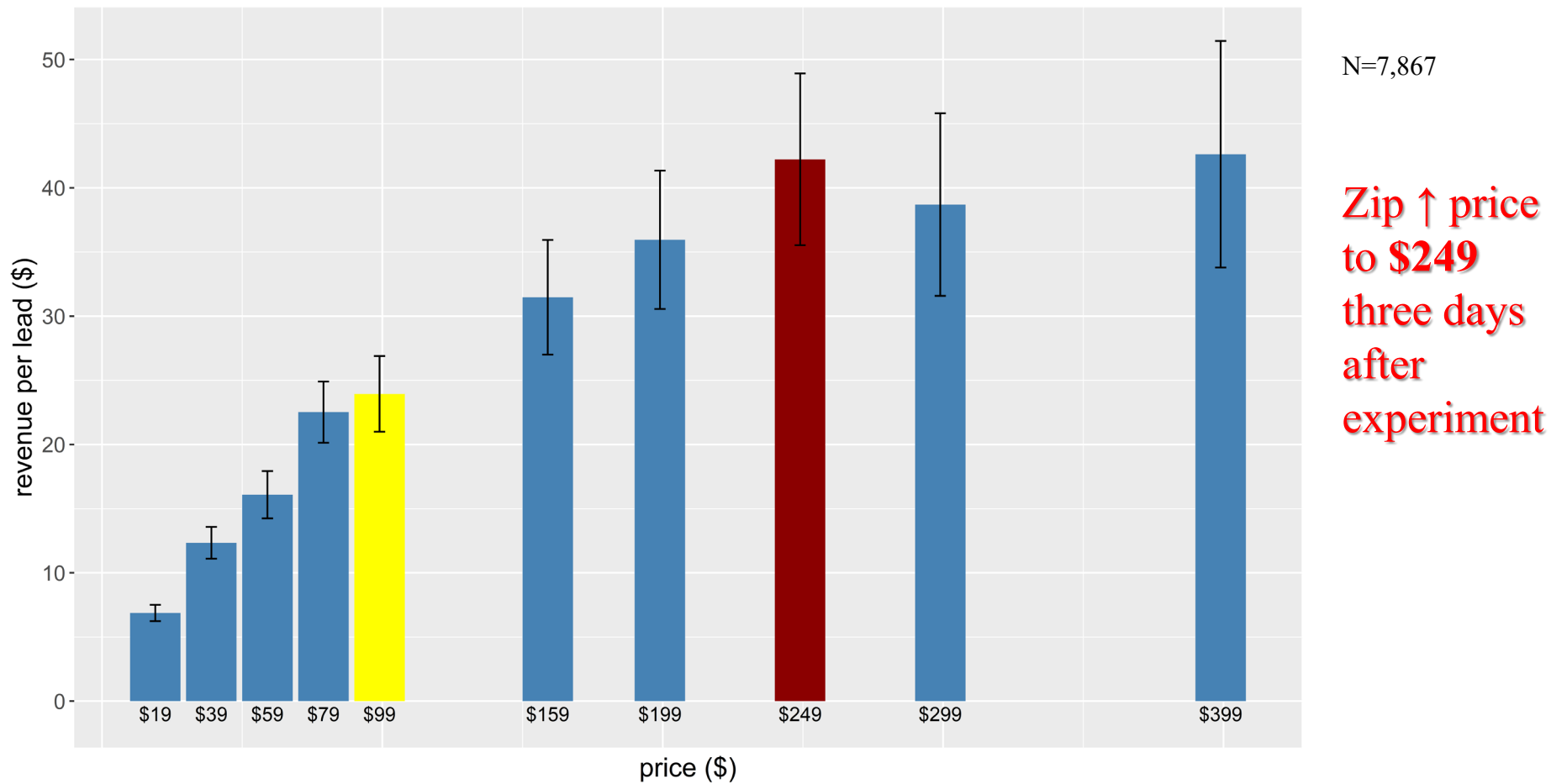
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The Ziprecruiter Price Experiment

Implementation: Price & revenue per lead

22

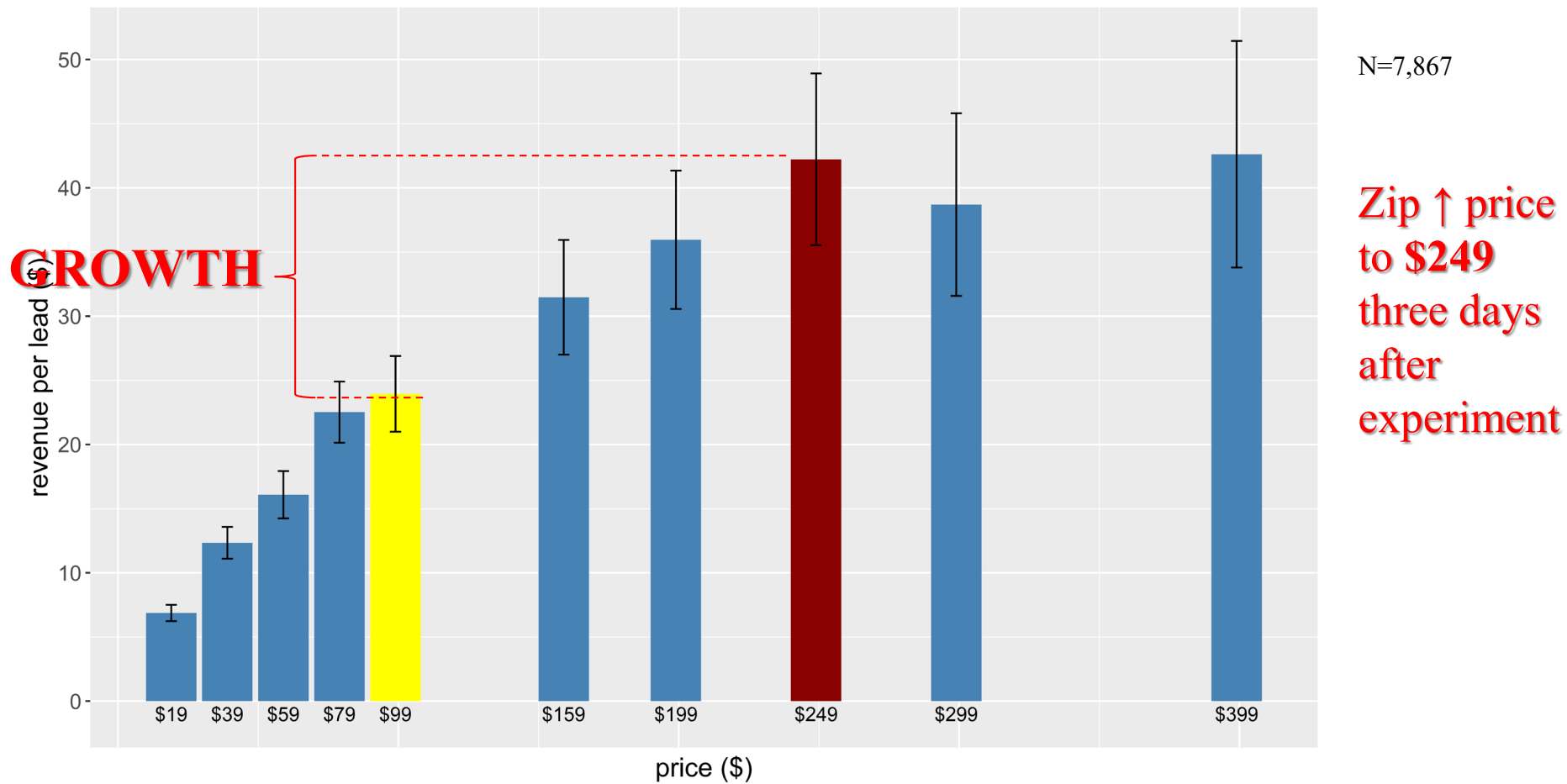


□ Even with model & data, need to use judgment to **implement** a price

The Ziprecruiter Price Experiment

Implementation: Price & revenue per lead

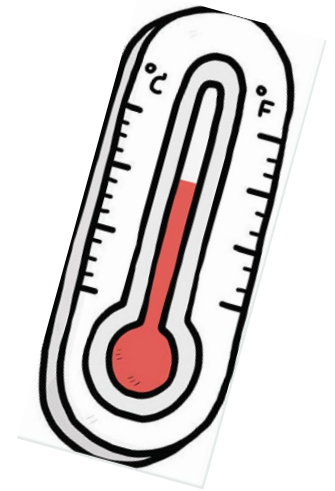
22



□ Even with model & data, need to use judgment to **implement** a price

The Ziprecruiter Price Experiment

Evaluation



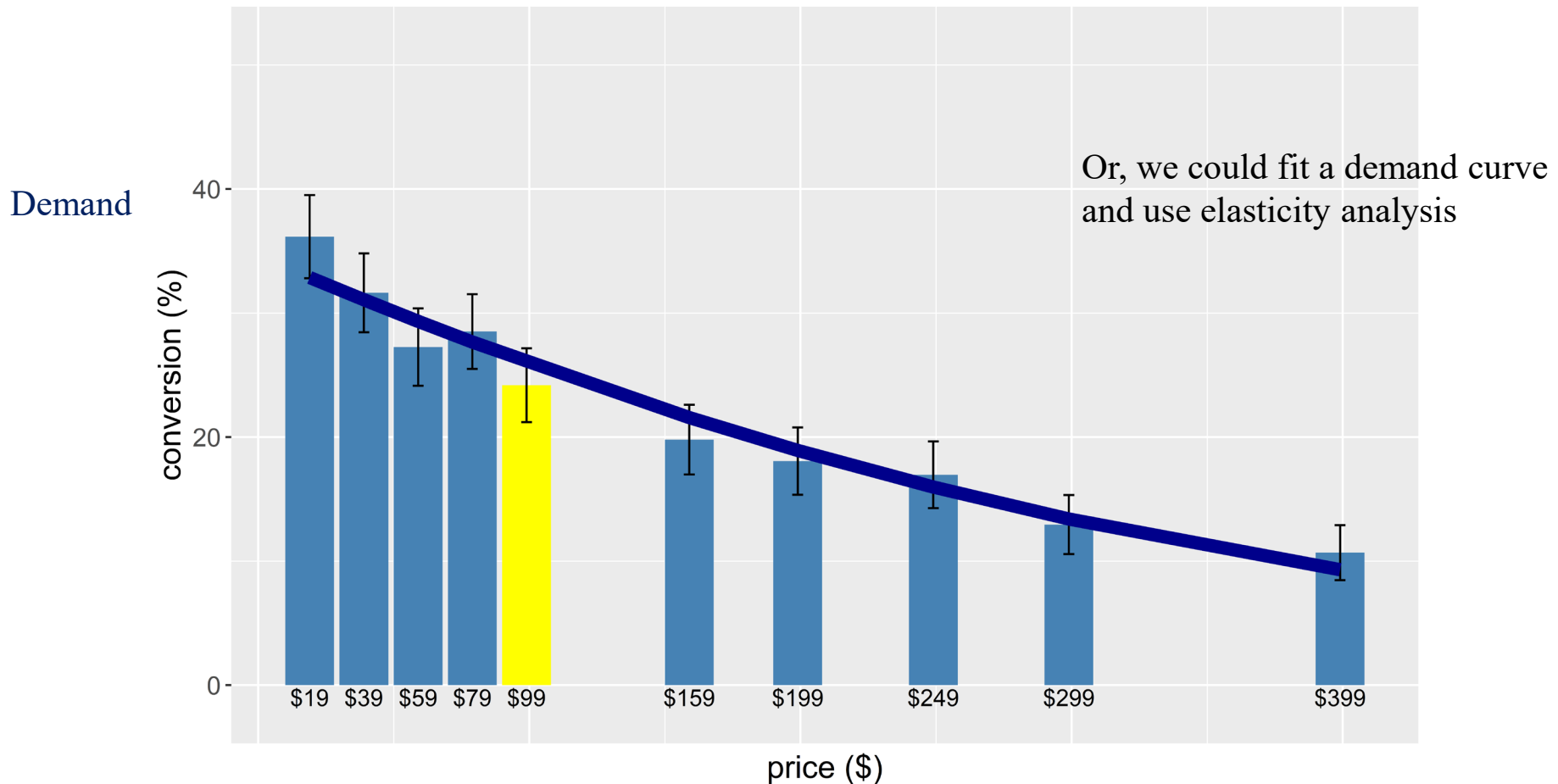
Optimization increased monthly profits by 65%

➡ Approximately \$70 million in **incremental** profits over 5 years!

Field Experiment to Measure Demand

Example: online recruiting service

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□ Fitted *logistic regression* (see appendix)

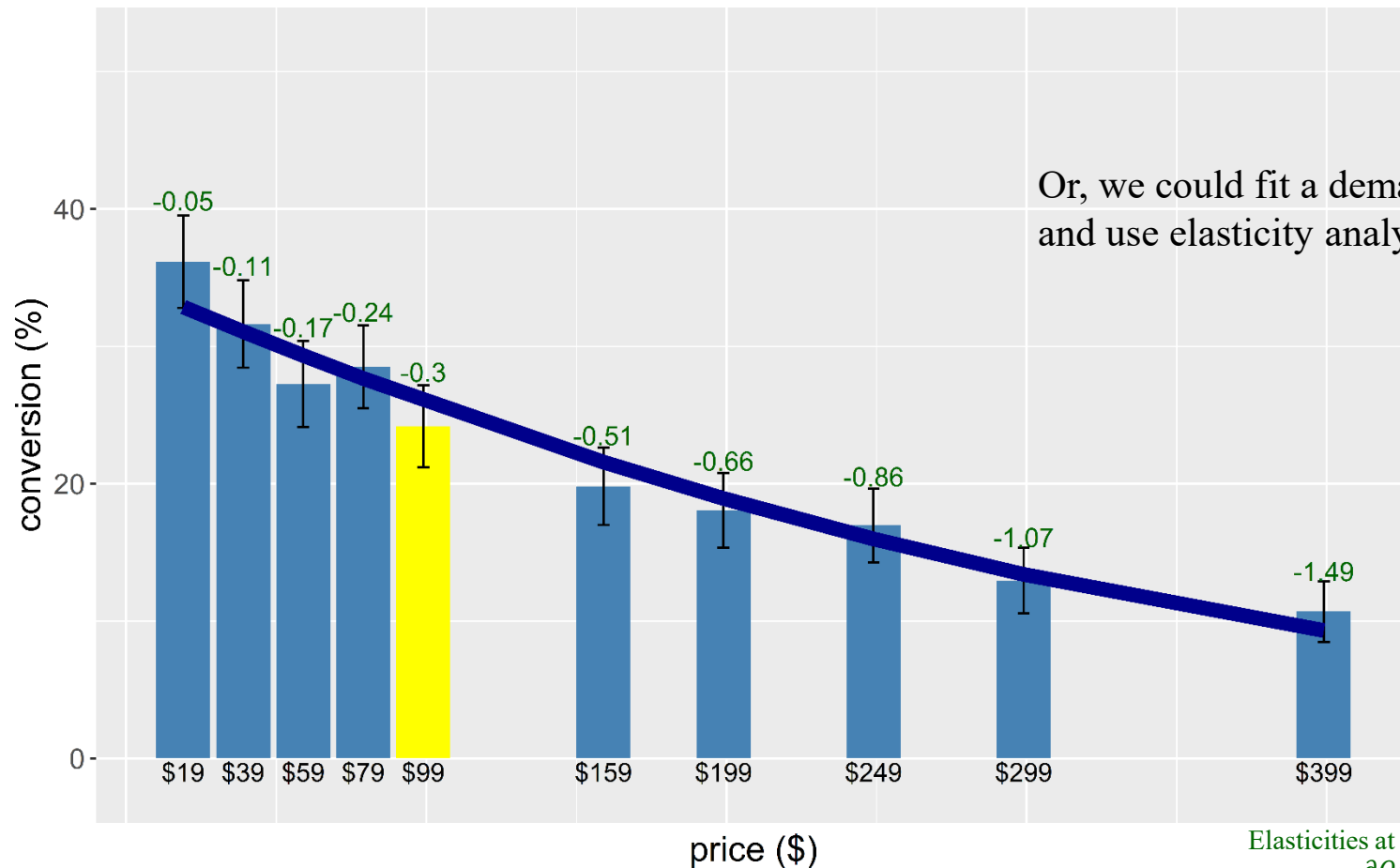
Source: [Dubé and Misra \(2023\)](#)

Field Experiment to Measure Demand

Example: online recruiting service

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Demand



□ Fitted *logistic regression* (see appendix)

Source: [Dubé and Misra \(2023\)](#)

Field Experiment to Measure Demand

Example: online recruiting service

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Demand



□ Fitted *logistic regression* (see appendix)

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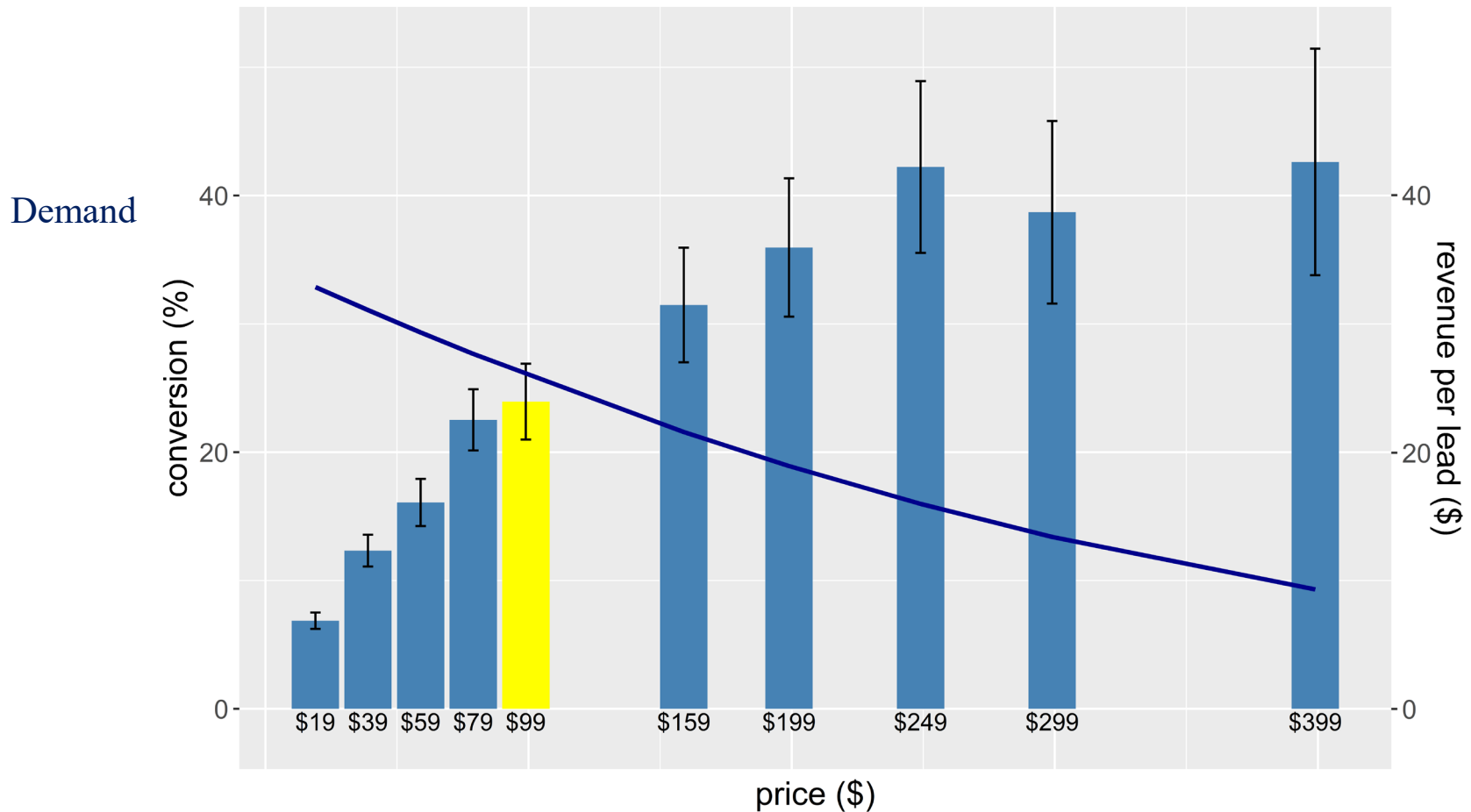
Elasticities at test prices:

$$\beta(p) = \frac{\partial Q(p)}{\partial p} \frac{p}{Q(p)}$$

Field Experiment to Measure Demand

Example: online recruiting service

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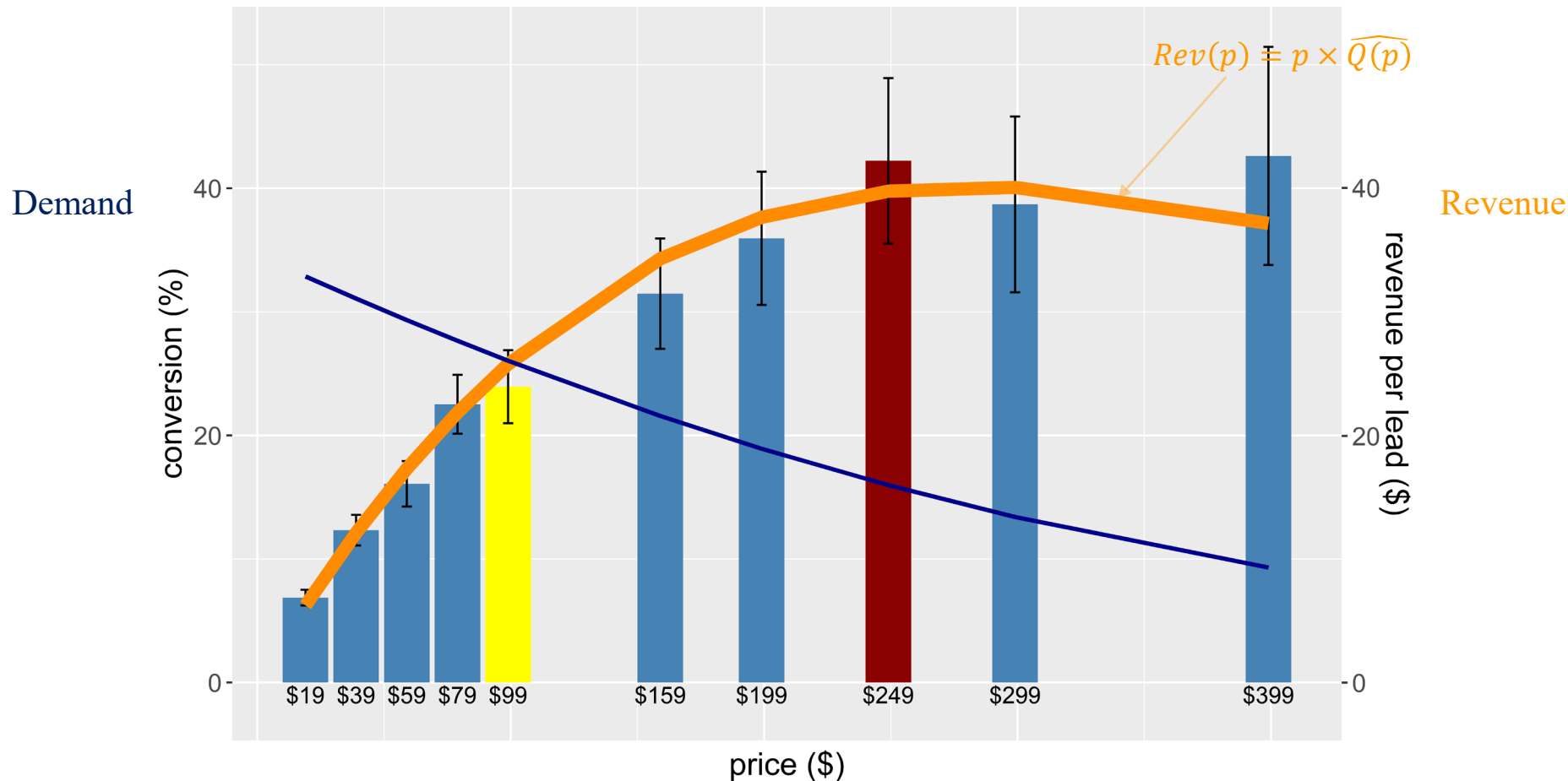
□ Now add experimental revenue data

Source: [Dubé and Misra \(2023\)](#)

Field Experiment to Measure Revenue

Example: online recruiting service

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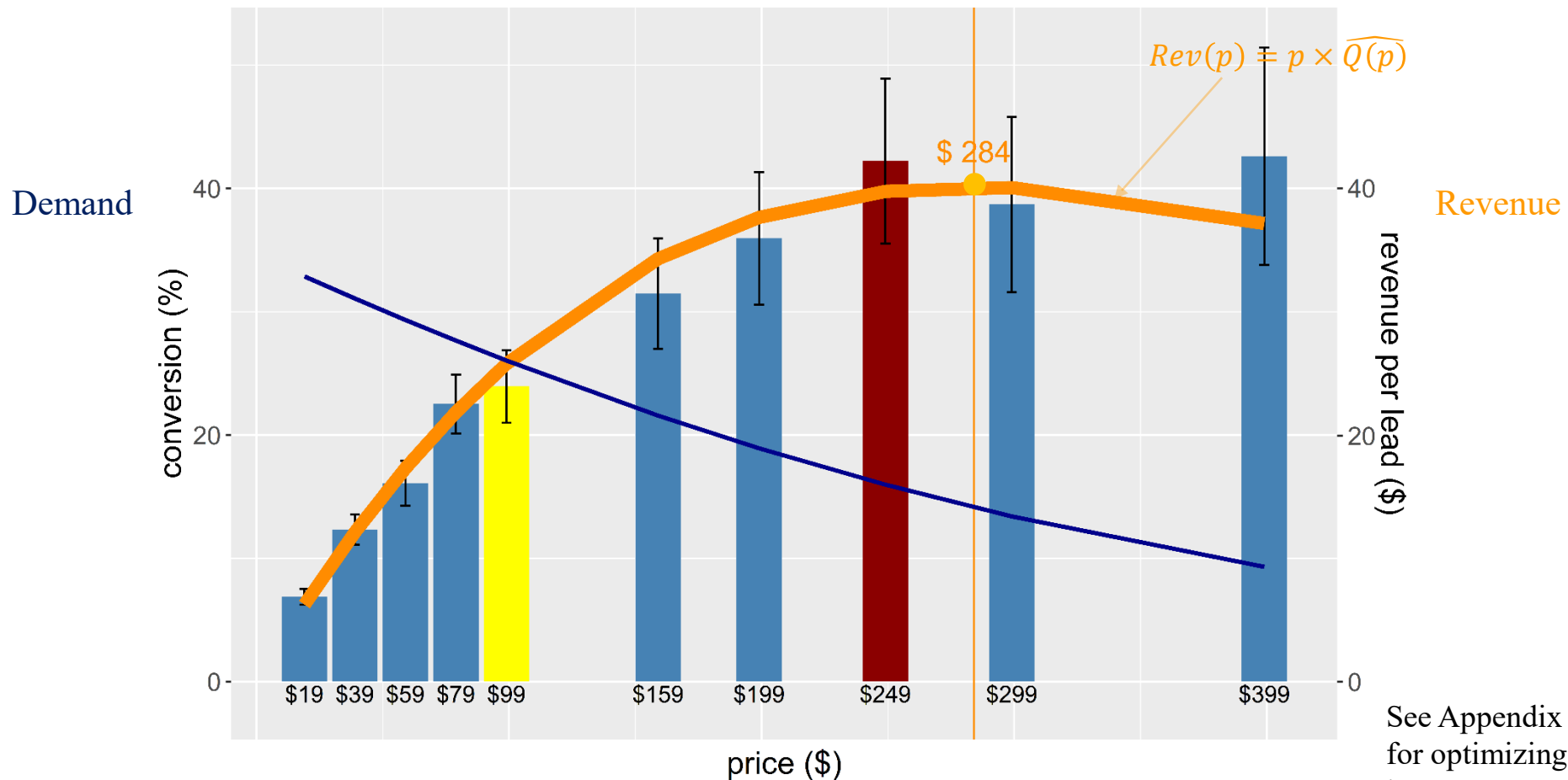


□ Predicted revenues based on fitted logistic demand

Field Experiment to Optimize Price

Example: online recruiting service

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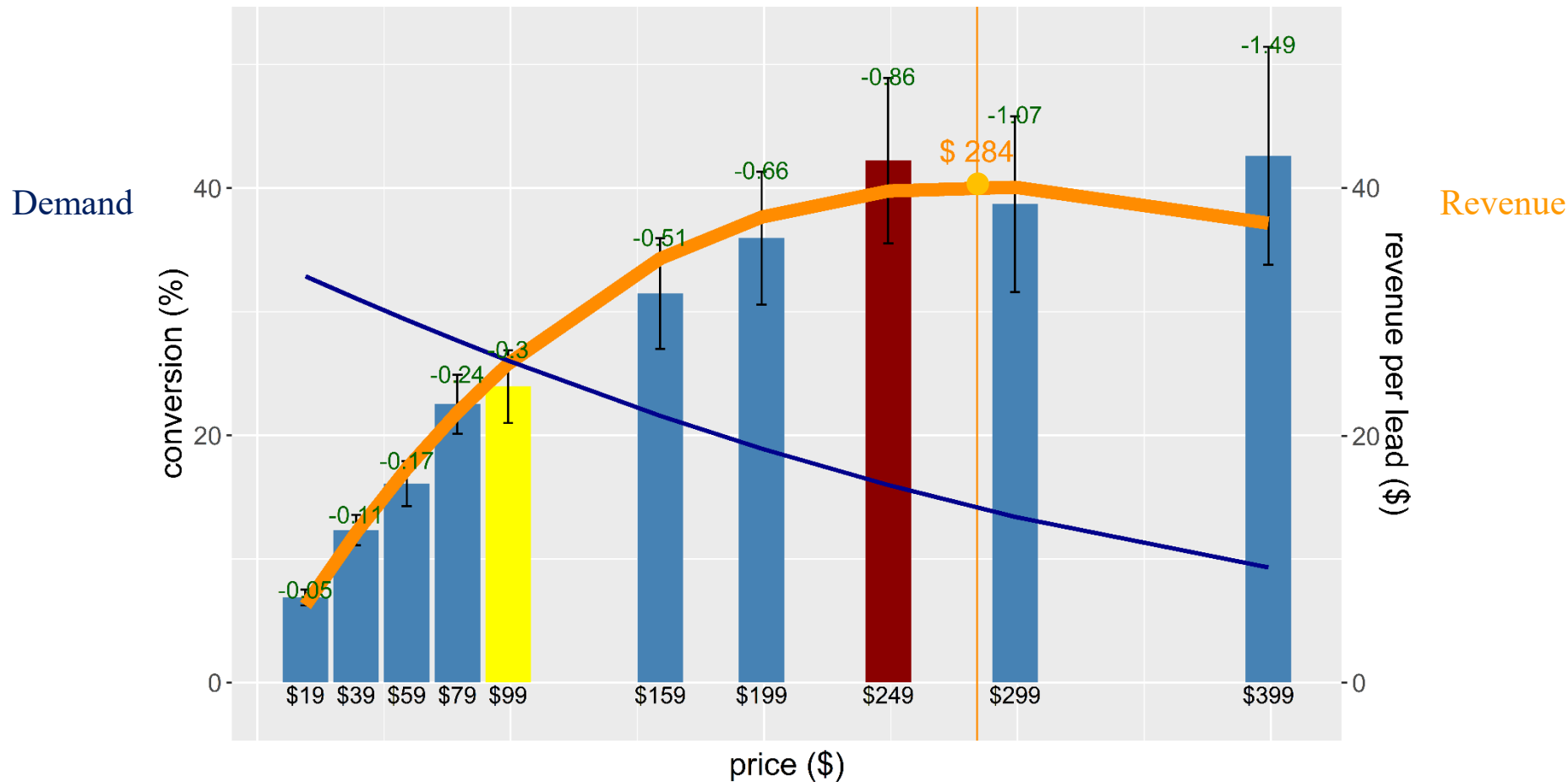
- Gains from optimized pricing
- Lower conversion BUT higher revenues

See Appendix
for optimizing
in R

Field Experiment to Optimize Price

Example: online recruiting service

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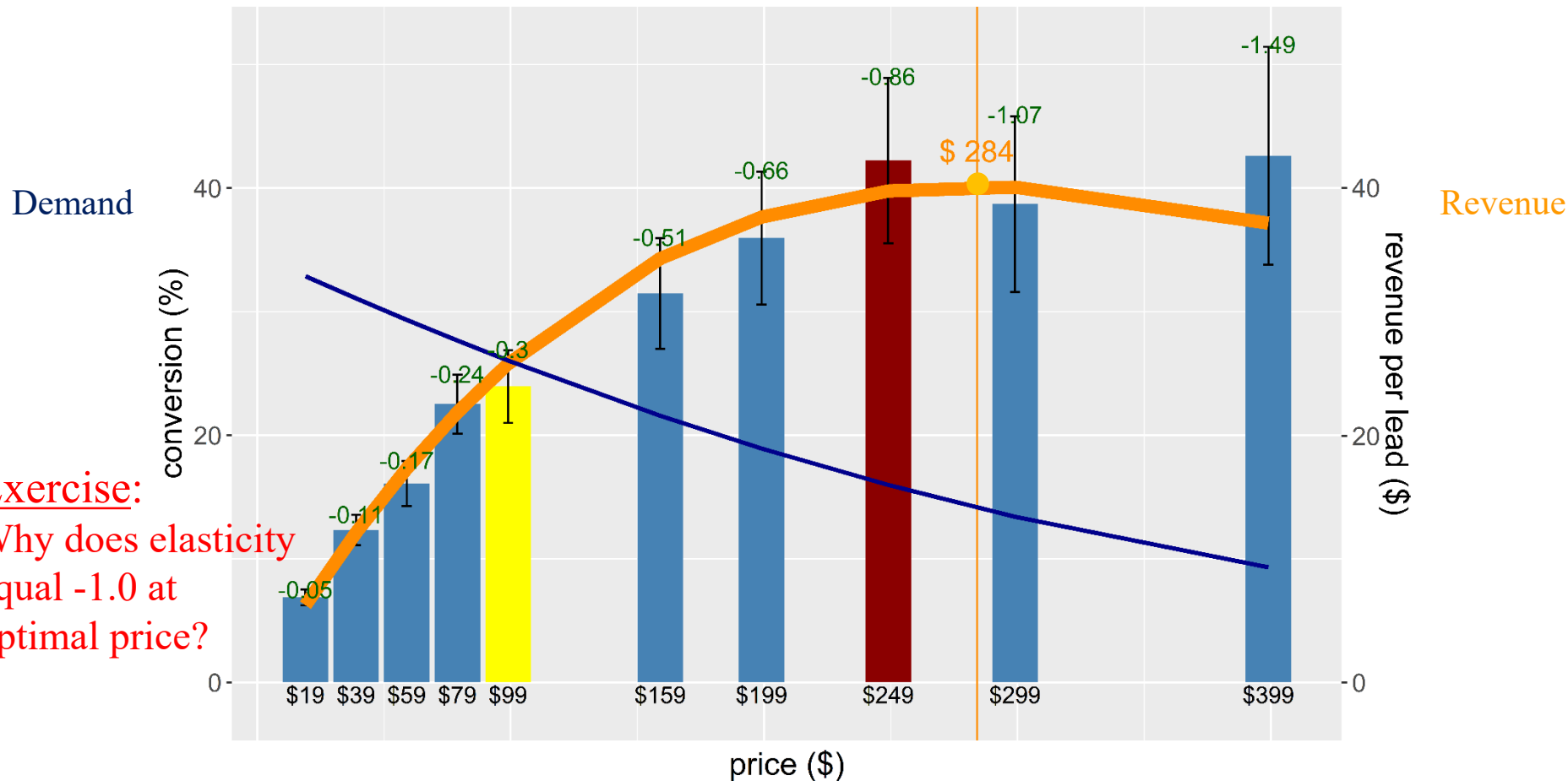


□ Let's add some demand elasticities ...

Field Experiment to Optimize Price

Example: online recruiting service

29

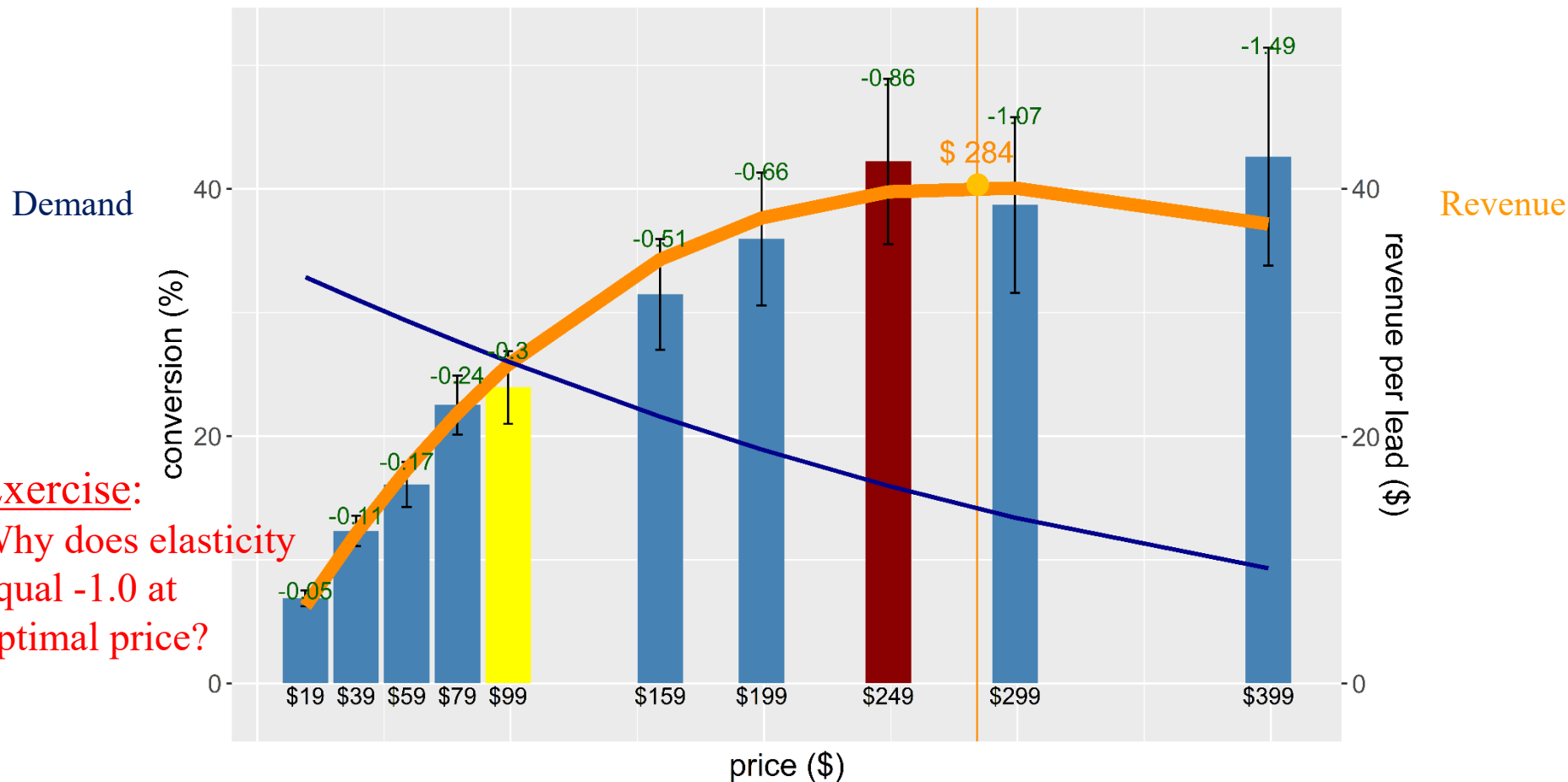


□ Let's add some demand elasticities ...

Field Experiment to Optimize Price

Example: online recruiting service

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- Let's add some demand elasticities ...
- See module 9 for pricing to the LTV of customer

Ziprecruiter Pricing

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- The **empirical strategy** for Ziprecruiter pricing:
 1. **Business Objective**: set price to maximize profit
 2. **Theory**: economic value to the customer
 3. **Model**: demand
 4. **Data & Method**: price experiment to estimate demand
 5. **Implementation**: use judgment & analytics to select price
- See extra mini-case for consumer goods pricing example

Take Away

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- Using the **Price Elasticity of Demand** for Pricing:
 - ▣ Typically want to price on elastic region of demand
 - ▣ Optimal pricing via ***inverse-elasticity rule***
- Elasticity measurement via randomized controlled experiment
- When “enough” prices are tested, you can trace out demand and revenues to locate “optimal” price

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Extra: Empirical Estimates of Price Elasticities

Interpretation & Generalizations

How do you know if your estimate is right?

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- Presence of closer **substitutes** implies higher elasticity
- Examples:
 - ▣ *Insulin*: canonical example of (totally?) inelastic demand
 - ▣ *Wheat*: canonical example of (perfectly) elastic demand

Marketing for Good: can price be too high?

Even Insulin exhibits price elasticity!!!

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Diabetes Patients at Risk From Rising Insulin Prices

A Yale study found that one in four patients admitted to cutting back on insulin use because of cost. The consequences can be deadly.

6-22-2018, *New York Times*

- No generic versions
- Between 2002 and 2013, price jumped from \$40 to \$130
- Survey found many (especially those ineligible for Medicaid and unable to afford good insurance) report under-using insulin and rationing.

Interpretation & Generalizations

How do you know if your estimate is right?

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- Average Elasticity for CPG (1,851 elasticities from 81 studies):
 - -2.62 ([Hanssens 2015](#))
- Small market share consumer brands often observed to be more price sensitive than large market share brands.
- We also saw how pricing theory can provide a lower bound on plausible elasticities if prices are **optimized**

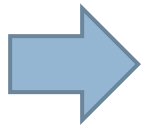
Elasticities and Margin Arithmetic



Margin Arithmetic

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- In practice, most firms do not know their demand.
- Break-even analysis generates useful margin arithmetic as a surrogate



You can transform a uninformed debate about whether to implement a tactical price cut (e.g., -5%) and re-focus it on price-sensitivity of demand.

- OK, you probably don't know the elasticity, but...
 1. The debate is now more focused
 2. You might have a sensible *gut instinct* about price-sensitivity (e.g., you might know that your product is at least a bit differentiated)
- Let's take a look at the margin arithmetic ...

Price cutting and price elasticity

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- Proposed price cut: $\frac{\Delta P}{P} \times 100 = -5\%$

- Minimum percent quantity increase to **break even**:

$$\begin{aligned}\Delta\pi &= (p_1 - c)Q_1 - (p_0 - c)Q_0 = \Delta P Q_0 + (p_1 - c)\Delta Q = 0 \\ \Rightarrow \frac{\Delta Q}{Q_0} \times 100 &= -\frac{\Delta P}{(P_1 - c)} \times 100\end{aligned}$$

- Example: Suppose firm currently has a 10% margin:

$$\begin{aligned}\frac{(P_0 - c)}{P_0} \times 100 &= 10\% \\ \Rightarrow c &= 0.9P_0\end{aligned}$$

- So break-even demand change is:

$$\frac{\Delta Q}{Q_0} \times 100 = -\frac{-0.05P_0}{(0.95P_0 - 0.9P_0)} \times 100 = 100\%$$

Price cutting and price elasticity

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- Many turn to price cutting as a means of stimulating sales.
Does this practice stimulate *Profits*?
- For low-margin products, it requires HUGE Elasticities!

Suppose we cut price by 5%, how much does volume have to increase to leave *profits unchanged* (i.e., to break even)?

$$\frac{\frac{€2600 - €53}{€2600} \times 100 = 98\% \rightarrow E = -1.02$$

Current Contribution Margin	Volume Increase Required	Price Elasticity Required
10%	100%	-20
25%	25%	-5
50%	11%	-2.2

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Ethical Margins?

Luxury Goods

An inconvenient truth about high margins at Dior & Armani

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- During a 2024 raid, Milan prosecutors discovered that:
 - ▣ Dior sells a handbag at €2,600 for which it paid the supplier only €53
 - This implies a contribution margin of $\frac{€2600 - €53}{€2600} \times 100 = 98\%$
 - If this is “optimal”, it implies an elasticity of $\beta = -1.02$
 - ▣ Armani sells a handbag at €1,800 for which it paid the supplier only €250
 - This implies a contribution margin of $\frac{€1800 - €250}{€1800} \times 100 = 86\%$
 - If this is “optimal”, it implies an elasticity of $\beta = -1.16$
- How are the costs so low? Exploitation of foreign labor

See: [“Raids Find Luxury Handbags Being Made by Exploited Workers in Italy,” by Nick Kostov, WSJ, 7-2-2024](#)

Ziprecruiter Data

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- Vector $\mathbf{y}$ has individual purchase decisions (0 or 1)
- Vector $\mathbf{p}$ has the price charged to each individual
- Data would look like the following:

Customer I.D.	Purchase (y)	Price (p)
1	1	\$99
2	0	\$99
3	0	\$139
4	0	\$259
5	1	\$99
6	0	\$79
7	1	\$79
...	...	...

A simulation in R to illustrate

Code to create simulated choice data

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- `N = 8000` # number respondents
- `betatrue = matrix(c(-0.64,-0.004))` # inter. & price coeff
- `testprices=matrix(c(19,39,59,79,99,159,199,249,299,399))`
- `Price = sample(testprices,size=N,replace=T)` # randomize prices
- `epsilon = -log(-log(runif(N))) + log(-log(runif(N)))` # draw logistic errors
- `U = betatrue[1]+betatrue[2]*Price+epsilon` # buying utility
- `Y = (U>0)*1` # choice data

Estimate the logistic regression in R

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- If you have the vectors **Y** and **P** in memory, you can estimate the logistic demand using the “*glm*” function in R
- Estimate the logit demand using built-in glm function in R:
 - `out = glm(Y~P,family=“binomial”)`
- Extract the demand coefficient estimates
 - `betahat = out$coefficients`

Predict Level of Logistic Demand in R

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- Predict Demand at a given price $\tilde{p}$:

$$S_Z(\tilde{p}) = \frac{1}{1 + \exp(-\text{beta}[1] - \text{beta}[2] * \tilde{p})}$$

Which is the probability that a customer buys at price $\tilde{p}$.

- `share = function(x,beta) 1/(1+exp(-beta[1]-beta[2]*x))`
- `Share.base = share(99,betahat)`

which is predicted share at \$99 using MLE results.

Now we can plot demand

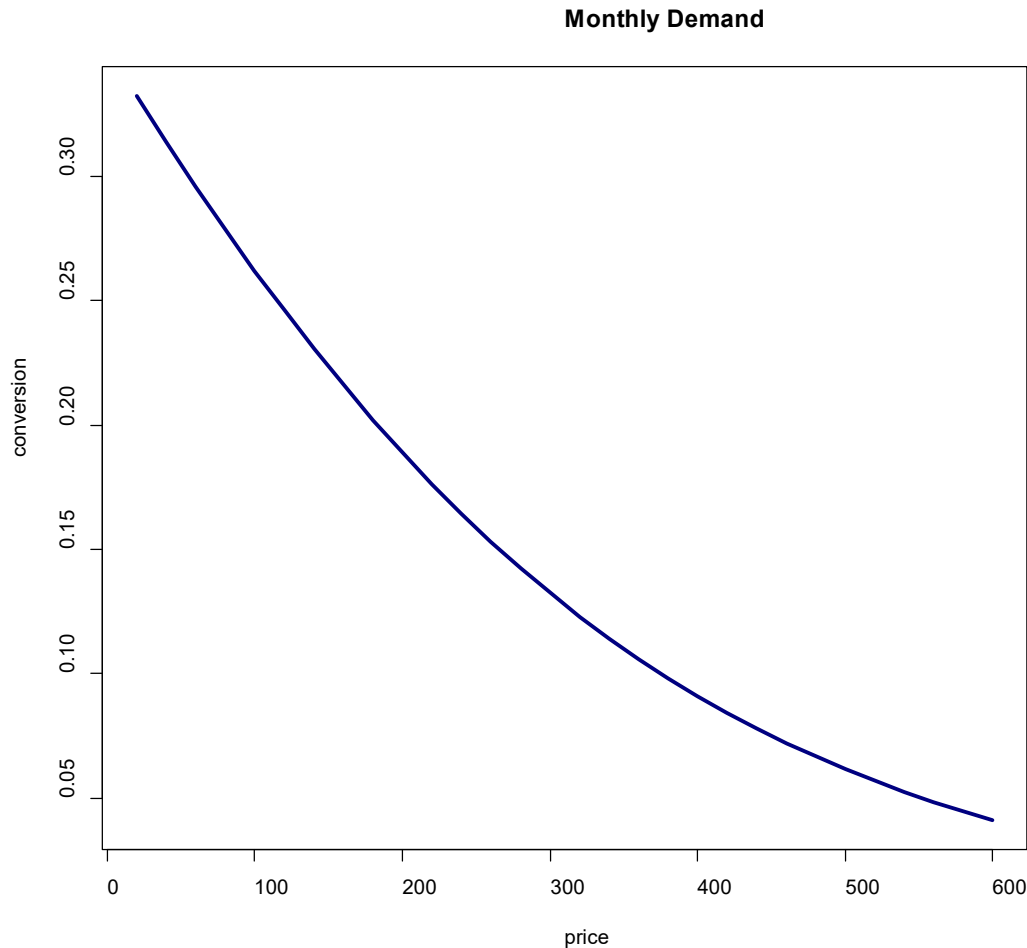
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- # Function to compute logit probability
- `share = function(x,beta){1/(1+exp(-beta[1]-beta[2]*x))}`
- # Plot Predicted Demand along a price grid
- `Pricegrid = seq(from=20,to=600,by=20)`
- `Qgrid = share(Pricegrid,betahat)`
- `plot(Pricegrid,Qgrid,main="Monthly Demand",type="n",ylab="conversion",xlab="monthly price ($)")`
- `lines(Pricegrid,Qgrid)`

A simulation in R to illustrate

Plot demand estimates

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Compute the Price elasticity of demand

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- Price Elasticity of Logit demand at given price $\tilde{p}$:

$$\begin{aligned}elas(\tilde{p}) &= \frac{\partial S_Z(\tilde{p})}{\partial p} \frac{\tilde{p}}{S_Z(\tilde{p})} \\ &= beta[2] * (1 - prob(\tilde{p})) * \tilde{p}\end{aligned}$$

- See if you can derive this result using calculus and a pen & paper.
- R code:
 - `Elas = function(x,beta)beta[2] * (1 - 1/(1+exp(-beta[1]-beta[2]*x)))*x`
 - `Elas.base = Elas(99,betahat)`

which is predicted elasticity at \$99 using MLE results.

Compute the Profit-Maximizing Price

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1. Construct a function in R to compute profits:

➤ `f = function(x,beta) 1/(1+exp(-beta[1]-beta[2]*x))*x`

2. Use the optimize command (I restrict price below \$1000):

➤ `out.opt = optimize(f,c(0,1000),betahat,maximum=T)`

3. Check elasticity at solution:

➤ `p.star = out.opt$maximum` # optimal price

➤ `elas.star = Elas(p.star,betahat)` # elasticity at the optimal price

Compute the Profit-Maximizing Price

You can also use the **inverse-elasticity rule!**

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- Function to measure difference in elasticity and “-1”:
 - `ElasRule = function(x,beta) {`
 - `beta[2] * (1 - 1 / (1 + exp(-beta[1] - beta[2]*x))) * x + 1`
 - `}`
- Load the *solver* function in R “nleqslv”
 - `install.packages("nleqslv")` `# install package`
 - `Library(nleqslv)` `# load library`
- now use solver to find optimal price (easy to do in Excel too):
 - `out = nleqslv(x=0,fn=ElasRule,beta=betahat)`
- Extract optimal price:
 - `p.star1 = out$x`